Payroll Tax Incidence: Evidence from Unemployment Insurance

by

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Abstract

Economic models assume that payroll tax burdens fall fully on workers, but where does tax incidence fall when taxes are firm-specific and time-varying? Unemployment insurance in the United States has the key feature of varying both across employers and over time, creating the potential for labor demand responses if tax costs cannot be fully passed through to worker wages. Using state policy changes and administrative data of matched employer-employee job spells, I study how employment and earnings respond to unexpected payroll tax increases for highly exposed employers. I find significant drops in employment growth driven by lower hiring, and minimal evidence of passthrough to earnings. The negative employment effects are strongest for young workers and single-establishment firms.

Keyword*:* unemployment insurance, labor demand, payroll taxation

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1 Introduction

Employer taxes on labor earnings are used to fund a variety of social insurance programs, and policymakers frequently view payroll tax cuts as a way to stimulate employment. In the theory of tax incidence, the resulting employment and wage change depends on the elasticities of labor demand and labor supply. Because labor demand is assumed to be much more elastic than labor supply, canonical models assume payroll tax costs are born entirely by workers, resulting in little to no effect on employment.[1] However, there may be imperfect pass-through in the presence of imperfect competition or downward wage rigidities, leading to a decline in employment rather than wages.[2] Hamermesh (1979) points out that if the entire burden of a payroll tax could be shifted onto wages, employers would not so vehemently oppose tax increases. Furthermore, payroll tax changes that only affect a subset of firms may prevent employers from fully passing costs through to wages.[3]

In this paper, I exploit quasi-random state unemployment insurance (UI) tax increases to estimate payroll tax incidence in the United States. The third largest payroll tax in the country, UI is an employer-specific and annually time-varying tax that is administered at the state level. Employer-specific tax rates are an increasing function of the amount of UI benefits paid out to previous employees, and in 2010 and 2011, state-legislated UI tax increases raised the payroll tax costs for highly exposed employers in multiple states. These temporary tax increases, which were uncorrelated with state economic conditions, ranged from $200 to $600 per worker annually – translating to average effective rate increases of 1.2% of payroll. I use administrative data from the US Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) to estimate the impact of these increases on employment growth and worker earnings.

[1] As evidenced by early estimates from Brittain (1971) and Gruber (1997).

[2] Using data from a large payroll processing firm, Grigsby et al. (2021) find very low rates of wage cuts, providing strong evidence for downward nominal wage rigidity. Using Census Bureau data, Murray (2021) estimates that downward nominal wage rigidity accounted for at least 23% of excess job destruction during the financial crisis of 2008.

[3] Likewise, Saez et al. (2012) and Saez et al. (2019) found that payroll tax changes only affecting a subset of workers within a firm also failed to result in full pass-through.

My research design exploits two dimensions of tax variation for identification: cross-state variation due to exogenous state policy changes, and within-state variation based on employer exposure. Highly exposed employers, defined as those with a history of recurring layoffs, are charged UI taxes close to the state maximum. Thus, when a state policy change increases the maximum UI taxes on employers, high-exposure employers face binding tax increases. Meanwhile, employers with no recent history of layoffs have minimal exposure to these tax increases because their tax rates are already far away from the maximum. I combine this additional dimension of variation into a difference-in-difference-in-differences (triple-difference) research design, comparing the outcomes of high- versus low-exposure employers located in states that legislated tax increases versus their counterparts located in states that did not. The inclusion of the third difference between high- and low-exposure employers allows us to net out potentially confounding state-level factors, such as local economic conditions or other state policies in the wake of the Great Recession.

I estimate negative impacts of state UI tax increases on employment growth, and no evidence of pass-through to earnings except for the lowest-earning workers (whose earnings declines could be driven by a reduction in hours rather than wages). For each 1% increase in a firm's effective tax rate, year-over-year employment growth declined by an average of 2.24 percentage points. Event study estimates found the largest effects in the 2nd and 3rd year after the initial policy changes, which suggest that these UI tax increases hindered the economic recovery after the Great Recession. I test whether this drop in employment growth is driven by increased separations or reduced hiring, and find evidence of the latter. The quarterly share of new hires fell by up to 0.7 percentage points for each 1% rate increase, while the effect on the share of separations was small and statistically insignificant. These findings align with firm tax incentives: Because low taxable earnings thresholds cause most workers to exceed their taxable payroll early in the year, additional separations provide no further UI tax savings whereas each new hire incurs an additional UI tax cost. Heterogeneity analysis shows that the negative employment effects are stronger for young (age under 25) workers,

and earnings effects – while still below full pass-through – were only detected for low-earning workers. Additionally, single-establishment firms were more responsive to tax increases than multi-establishment firms, suggesting that cash constraints and internal networks in the face of an unexpected tax increase may play an important role. Taken together, these findings suggest that the UI payroll tax increases following the Great Recession played a role in the so-called "jobless" recovery.

The recent literature on payroll tax incidence has combined administrative microdata with exogenous policy changes to provide strong evidence of imperfect pass-through. These studies largely focus on European and South American countries, as the lack of variation in federal payroll taxes has caused U.S.-based studies to be absent from the recent literature. The first contribution of this paper is to provide a U.S. context to study payroll tax incidence, because differences in labor market institutions may prevent the estimates of tax incidence in one country from generalizing to another. Despite the U.S.'s low rates of unionization, my results are consistent with the existing literature in showing minimal pass-through to earnings, and significant employment responses instead. I estimate large labor demand elasticities consistent with Ku et al. (2020), which studied a similar setting of regional tax increases in Norway. The small existing literature on UI tax incidence in the United States either relied on cross-sectional data (Anderson and Meyer (1997)) or variation within a single state.[4] This paper contributes to the literature by using exogenous policy changes from multiple states for identification, and by estimating additional heterogeneity by worker and employer characteristics. My finding that only low-earning workers experienced earnings declines from UI tax increases suggests that UI tax incidence is more regressive than previously thought.

The second contribution is the study of an unexpected and firm-specific payroll tax increase. Most existing studies exploit policy changes that led to payroll tax cuts, but due to

[4]Anderson and Meyer (2000) studies the introduction of experience rating in Washington, and found pass-through of long-run industry tax costs but not employer-specific rates. Using administrative data from Florida, Johnston (2021) found little effect of annual changes in employer-specific rates on worker earnings, but large impacts on hiring.

the presence of cash constraints, tax hikes may have an asymmetric and particularly strong effect on firm behavior. For example, Saez et al. (2019) found relatively modest employment gains from a payroll tax cut on young workers, and Fenizia et al. (2024) found no employment effect from a payroll tax reduction targeted towards apprenticeships at small firms. Furthermore, firms that have recently laid off workers are the most likely to be cash constrained, and the state UI tax increases I study directly affected this subset of the firm population. Benzarti and Harju (2021*a*) found that a payroll tax cut in Finland helped treated firms weather the Great Recession despite having minimal impact on employment beforehand, suggesting a cyclical component to firm responses. My study of the period directly following the Great Recession provides further evidence that firms are more responsive on the employment margin when they are cash constrained, and provides a potential explanation for the slow recovery following the Great Recession.

Another contribution is the estimation of heterogeneous effects across worker and employer characteristics. Benzarti and Harju (2021*b*) found that firms with greater payroll taxes substituted away from low-skilled and manual workers. I find greater drops in labor demand for young workers, but not for those with less education. My finding that single-establishment firms were especially likely to decrease hiring in response to higher UI taxes, is consistent with the finding in Lobel (2021) that small firms were more responsive to payroll tax cuts. Biró et al. (2022) estimate heterogeneity in payroll tax incidence by firm productivity, finding that lower productivity firms experienced employment responses while higher productivity firms passed tax cuts on to workers. While I do not observe firm productivity, my sample of high exposure employers likely overlaps with lower productivity firms, as they suffered layoffs prior to the tax increases. The large magnitude of my estimated labor demand elasticities imply that employers most likely to be exposed to payroll tax increases are also the ones least likely to be able to pass the costs on to workers. Thus, targeted payroll tax reductions towards these firms may actually be the most effective in stimulating employment.

The next section provides institutional details about U.S. Unemployment Insurance and describes the research design. Section 3 discusses the data and construction of my analysis sample. Section 4 presents employer-level results, and Section 5 presents a worker-level analysis of the incidence on earnings. Section 6 concludes.

2 Background and Research Design

2.1 Institutional Details

Unemployment insurance in the U.S. is funded solely through payroll taxes on employers.[5] Unlike Social Security or Medicare taxes, which are uniform across all employers, UI is experience-rated and therefore employer-specific. UI tax revenues reached a height of 1 percent of total wages in 2012, but this masks considerable heterogeneity across employers. Employers that lay off workers who then claim UI benefits are assigned a higher future tax rate - up to a statutory maximum.[6] These employer-specific rates are an increasing function of the dollar value of UI benefit claims relative to taxable payroll, and are calculated annually.[7] Appendix Figure A.2 plots total UI contributions over time as a percentage of total wages, and reveals the cyclical pattern of UI tax collections, where tax rates are highest in the years following recessions. Due to this time-varying nature of UI tax rates, businesses may have more difficulty passing the costs on to their workers, and changes in UI tax liabilities have the potential to influence employment as well as earnings.

Another unique aspect of UI is the low tax base in most states. The federally mandated level of taxable earnings is only $7,000, and because most states fail to index UI taxes to

[5] With the exception of New Jersey, which also charges workers a uniform payroll tax of 0.5%.

[6] Guo and Johnston (2021) provides a brief history of U.S. unemployment insurance, which was modeled after the experience rating in workers' compensation. Since its inception, policymakers have recognized the tradeoff between disincentivizing layoffs, and the cyclical tax burdens that dynamic experience rating would create.

[7] Appendix Figure A.1 illustrates a sample tax schedule from the state of Florida, where the maximum tax rate is capped at 5.4%. In 2008, only 5 percent of eligible employers were assigned the maximum tax rate, while 77 percent were at the minimum. However, by 2011 these shares changed to 22 percent and 63 percent, respectively, due to layoffs during the Great Recession.

wage growth, they have experienced a gradual erosion of the UI tax base over the last half-century. In 2009, the year before the first UI tax increases, the median UI tax base was only $10,000 – with the median state's taxable wages covering only 28% of total payroll. When workers' annual earnings exceed the UI tax base, UI becomes a lump-sum tax, independent of earnings. For example, an employer facing a UI tax rate of 5.4% and a tax base of $10,000 would contribute payroll taxes of .054*10,000 = $540 per worker for any worker earning more than $10,000 annually (see Appendix Figure A.3). Thus UI taxation is regressive in the sense that seasonal, part-time, and low-wage workers are taxed at a higher effective tax rate (as a share of earnings), and these types of workers become relatively more costly for a business facing high UI tax rates.[8]

Because unemployment insurance is administered at the state level, there is substantial variation in UI tax schedules across states. A tax increase in one state raises payroll costs for incumbent employers, without affecting tax regimes in any other. Many state UI programs are underfunded, leading them to deplete their UI trust funds during recessions. During the Great Recession, 34 states depleted their UI trust funds and were forced to take out loans from the federal government; this occurred again for 18 states during the height of the COVID-19 pandemic.[9] Although states recognized the need to replenish their trust funds following the Great Recession, policy responses varied greatly, with a total of sixteen states legislating UI tax increases, while other similarly indebted states chose inaction. During this period Congress attemped to pass the Unemployment Insurance Solvency Act of 2011, which would have raised the federally mandated UI tax base from $7,000 to $15,000, and paused interest payments on state UI loans until 2013. Had the legislation passed, individual states would have no need to impose their own tax increases. Therefore, the fact that some states

[8]Huang (2022) finds that higher state taxable wage bases increases teenage employment rates, and Duggan et al. (2022) finds that higher tax bases increase labor demand for low-wage part-time workers.

[9]When state trust funds borrow from the federal government to pay out UI benefits, they are given two years to repay that debt. After this window, every single employer is charged $21 per worker per year until the loans are paid back. These consequences are akin to a tax increase for employers, but the amount is small and spread out over the entire state (whereas state-legislated tax increases would charge high-exposure employers significantly more).

chose to wait for federal action produced exogenous variation in state policy changes, despite similar financing needs.[10]

2.2 Conceptual Framework

As an employer-side payroll tax, employers are charged UI taxes for each worker employed, and these are collected regardless of profitability. The cyclicality of experience rating – tax rates rise in the years following layoffs – also creates a large potential tax burden for firms in years when they may be most cash constrained. This is likely to generate especially large labor demand elasticities in response to UI tax increases.

Furthermore, the low base of taxable wages causes UI to become a per-capita labor cost rather than a tax that scales with earnings. This also imposes an uneven distribution of tax burdens throughout the calendar year, where tax payments are highest in Q1 when all earnings are still under the taxable wage base, and fall throughout the year as earnings gradually exceed the UI tax base. Table 1 provides an illustration of this phenomenon amongst the treatment and control states in my analysis sample by focusing on a high-tax sector such as construction to proxy for high-exposure employers. UI tax contributions are calculated by dividing quarterly UI contributions by monthly employment, representing the industry's average per-worker cost in each quarter. In 2011, a year when experience rating was at its peak, effective UI tax rates were 3-4% in Q1 and fell gradually over the course of the year. This also means that for a given unit of labor demand at the end of the calendar year, a new hire is costlier for the firm than an existing worker. The existing worker's marginal UI tax cost is zero, whereas a new hire would still be charged the full UI tax cost because their earnings are still below the UI tax base.

[10] Additional details about state UI tax increases are discussed in appendix subsection A.1.1.

2.3 Identification

This paper relies on a difference-in-difference-in-differences (triple-difference) research design to identify the causal impact of state UI tax increases. The difference-in-difference compares states that implemented policy changes to ones that did not, and a third difference compares high- and low-exposure employers within the same state. Because policy changes were only binding for employers near the maximum tax rate, we can use employers with minimal layoffs – and therefore low exposure to UI tax increases – to control for confounding state-level variation that may influence both tax policy and employment outcomes.

My research dataset covers 23 states and the District of Columbia.[11] Figure 1 graphs the distribution of maximum UI tax increases (tax base * maximum rate) from 2009 to 2011, against baseline 2009 tax levels. I exclude seven states from my analysis due to baseline differences in UI tax regimes.[12] From the remaining states, nine treatment states experienced statutory maximum UI taxes increases of more than $100 from 2009 to 2011, and the remaining eight control states experienced minimal tax changes during this window. From 2009 to 2011, treatment states experienced per-capita maximum UI tax increases ranging from just under $200 to over $600, and the average increase was $378. In percentage terms, maximums increased by 67% on average, and the largest tax increase was experienced by South Carolina (in both level and percentage terms).[13] Figure 2 plots the locations of these 17 included states, and the change in maximum UI taxes from 2009 to 2011.

Appendix subsection A.1.1 provides more details about the quasi-randomness of state UI tax increases, which were sudden and unpredictable. To test the exogeneity of these tax increases, Figure 3 graphs four scatterplots comparing each state's UI tax increase with

[11]The states I have access to include: Arizona, Arkansas, California, Colorado, Delaware, District of Columbia, Florida, Illinois, Indiana, Iowa, Kansas, Maine, Maryland, Missouri, Montana, Nevada, New Mexico, Oklahoma, Oregon, Pennsylvania, South Carolina, Tennessee, Washington, and West Virginia.

[12]Six of these excluded states automatically index UI tax bases to wage growth, resulting in incremental and predictable tax increases every year throughout my sample period (and thus greater tax costs and trust fund solvency at baseline than the treatment states). I also exclude Pennsylvania, which experienced a UI tax *decrease* in 2011-12.

[13]Appendix Figure A.4 plots maximum and average UI taxes over time for each control state, and appendix Figure A.5 and Figure A.6 plots them for treatment and excluded states, respectively.

measures related to state economic conditions. Tax increases are not correlated with state unemployment rates, UI claim rates, UI benefit generosity, or UI trust fund loans. Although South Carolina experienced both high unemployment and the largest tax increase, other states heavily impacted by the Great Recession - such as California and Florida - chose not to increase UI maximums at all despite experiencing high unemployment and having insolvent UI trust funds. If it were the case that states hardest hit by the Great Recession were also the ones that chose not to legislate UI tax increases, this would also bias us against finding a treatment effect.

In order to focus on employers for whom state UI tax increases would be binding, I create a sample of high-exposure employers that are likely to be close to the maximum tax rate at the time of the tax increases. This is based on imputed layoffs, and details are described in subsection 3.1. This not only ensures that the treatment will be binding, but also has the added benefit of restricting to employers who already regularly separate workers into nonemployment. Thus, we avoid the typical endogeneity concern of tax rates being correlated with layoff rates (in the absence of a policy change), and the identifying assumption is that in the absence of UI tax policy changes, highly exposed employers in treatment states and control states would have evolved similarly in the years following the Great Recession.

Given the potential concern about differential state-level exposure to the Great Recession, I control for state-specific factors by further comparing the main sample of high-exposure employers to a placebo sample of employers that experienced minimal layoffs – the third difference in my triple-difference design. Employers that do not lay off workers will be assigned UI tax rates close to the minimum of the tax schedule. When a state increases their maximum UI tax (whether through a base increase or a change in the maximum tax rate), employers near the minimum will experience little to no change in their UI tax costs. We can thus use these low-exposure employers to control for confounding differences across treatment and control states that may influence employment outcomes.

It is also worth noting that after the Great Recession, there were significant federal extensions of UI benefit duration beyond the typical 26 weeks, through both the Emergency Unemployment Compensation (EUC) program and the Extended Benefits (EB) program. Benefit duration reached a potential maximum of 99 weeks in 2010-2011, before dropping in 2012 and coming abruptly to an end at the close of 2013. Farber and Valletta (2015) and Farber et al. (2015) found little impact of these benefit extensions on job finding, as they primarily caused fewer workers to leave the labor force and led to long-term unemployment. I assume the extension of UI benefits does not differentially impact worker labor supply during my analysis period, and Figure A.8 shows no correlation between UI benefit extensions and state UI tax increases.

3 Data

This project uses the Census Bureau's Longitudinal Employer-Household Dynamics (LEHD), an administrative employer-employee matched dataset of quarterly earnings, directly sourced from state UI records. Each employer in the LEHD is assigned a state employer identification number (SEIN), and this serves as the definition of a firm; this is also the level at which UI tax rates are assigned to employers. One employer can have multiple establishments within a state, but no employment outside of the state is counted. Large firms that have multiple EINs associated with their business may also own multiple SEINs within a given state, but each will be counted as a separate employer.

My analysis sample consists of employers from 9 treatment states and 8 control states, that were in operation from 2006 to 2009. I then restrict my analysis window to the period 2008 to 2013, to allow for at least two years of pre-change data and three years of post-change data. The inclusion of any years beyond this window of time would introduce additional state-level tax changes that could result in leakage across the treatment and control groups. I drop the Retail Trade and Accommodation & Food Services sectors (NAICS 44-45 and

NAICS 72), because these two sectors experience high turnover and low UI take-up rates such that it is very difficult to classify employers into high versus low exposure to UI taxes. The exclusion of these sectors should not affect the generalizability of my results, as they rank among the sectors with the lowest average UI tax costs, just above Healthcare and Education (Guo and Johnston (2021)).

The main outcome of interest, employment growth, is calculated year-over-year using the midpoint method, defined as $\frac{Emp_t - Emp_{t-1}}{\frac{1}{2}(Emp_t + Emp_{t-1})}$. To avoid seasonal patterns in labor demand, this measure compares employment in the current quarter to the same quarter of the prior year. I also count the number of new hires and separations in each quarter based on observed employment spells, and divide by employment to define quarterly shares of new hires and separations. Finally, I measure quarterly earnings at the firm level by taking the average over stable workers (which excludes new hires and separations).

3.1 Calculating UI Tax Exposure

State UI tax increases raised the maximum tax rate and/or UI tax base, both of which require a sufficiently high experience rating to be binding. Because I do not directly observe employers' payroll tax rates in the LEHD, I impute exposure to UI tax increases based on separation history. Employers who rarely separate workers into non-employment are inferred to have a low UI tax rate (low-exposure), and employers who separate a significant share of workers into non-employment will be inferred to have a tax rate close to the state maximum (high-exposure).[14] Details of this imputation are described in appendix subsection A.1.2.

For each employer, I aggregate the number of imputed layoffs over two separate pre-periods, and use these two cumulative shares to define inclusion in my analysis sample as a high-exposure firm. The first pre-period runs from 2006:Q1 to 2007:Q4, and the second runs from 2008:Q1 to 2009:Q2 (the Great Recession began in December 2007 and ended in

[14]In practice, employer-specific UI tax rates are determined by a formula that varies across states. Some states use a temporary look-back window of 3-5 years when assigning employer tax rates, while other states factor in employers' entire history of UI benefit claims and contributions.

June 2009).[15] High-exposure employers are defined as those that experienced total layoffs of 33-100% of their 2009:Q3 baseline employment in each of the two pre-periods. The purpose of using two separate pre-periods rather than the cumulative total is to identify firms with persistent UI risk, rather than firms that experienced a single mass layoff event during the Great Recession.

I also identify a subset of low-exposure empoyers, defined as those with cumulative imputed layoffs of less than 15% from 2006:Q1-2007:Q4, and less than 10% from 2008:Q1-2009:Q2. Given the likelihood of overestimating layoffs, an employer with an imputed layoff rate less than 10% very likely experienced zero or very little UI benefit claims. This set of employers, which are expected to be the least affected by UI tax increases due to their distance from the maximum tax rate, will act as a "placebo" low-exposure sample in my triple-difference research design. The cutoffs used to define high- and low- exposure employers were chosen to create roughly equal shares of each across the firm population, and each subgroup constitutes roughly 13% of the overall firm population in 2009:Q3.

I then restrict my analysis sample to firms employing between 20 and 500 workers at baseline (2009:Q3), to minimize measurement error in this imputation.[16] While the large majority of firms are small, the tax exposure of small firms is difficult to impute because whether a given layoff results in UI claims will make a larger difference the smaller the firm is.[17] On the other end of the spectrum, I exclude very large firms because of the potential for manipulating UI take-up. Lachowska et al. (2023) finds a positive correlation between employer size and UI appeals, and there exists an industry that specifically helps large

[15] I choose the 2006:Q1-2009:Q2 window in order to capture layoffs that could lead to binding UI tax increases post-policy change, because mass layoffs translate into higher employer-specific tax rates with a multi-year delay. For example, layoffs during the 2008-09 Great Recession caused tax rates to peak in 2011-12. UI claims that occurred prior to 2006 are unlikely to impact tax rates in 2010 and later, because firms will have already experienced tax increases by then. UI claims that occurred after the first half of 2009 are not even included in the formulas used to calculate UI tax rates until the 2011 tax year.

[16] The 2009 County Business Patterns estimates that nationally, firms between 20-500 workers made up 10% of firms, and 31% of employment.

[17] For example, expanding the analysis sample to include firms employing 10-19 workers almost doubles the sample size, and results in an attenuation of the estimated treatment effects.

employers contest UI claims.[18]

3.2 Summary Statistics

Table 2 reports summary statistics at baseline in 2009:Q3. The leftmost columns compare high-exposure firms across treatment and control states, while the rightmost columns compare low-exposure firms across treatment and control states.

Firm size is very similar across the high-exposure subgroup, with average employment of roughly 67 workers across treatment and control states. Low-exposure firms skew slightly larger, especially in treatment states. Imputed layoff shares match extremely well across treatment and control states, with the average high-exposure firm having experienced cumulative layoffs in 2006-07 that amounted to 57% of their 2009:Q3 employment, and 50% for the period of 2008:Q1 to 2009:Q2. This helps to alleviate the concern that treatment states might have more highly exposed employers than control states.

There is significant mismatch, however, in annual earnings, with control states paying higher wages than treatment states. This is likely due to state-level differences in cost-of-living and minimum wages, as we observe a similar gap within low-exposure firms. Federal mininum wage increases occurred in July of 2007, 2008, and 2009, and treatment states were more likely to have faced binding minimum wage increases.[19] Therefore, I include a control for the state minimum wage in all regression specifications. Additionally, the estimation of a triple-difference design comparing high- and low-exposure employers within the same state will also control for wage-related confounders across treatment and control states. While low-exposure firms also have significantly higher average earnings than high-exposure firms, this is primarily due to differences in turnover rates, as evidenced by the higher shares of new hires and separations among the high-exposure subgroup.

[18] "Contesting Jobless Claims Becomes a Boom Industry": `https://www.nytimes.com/2010/04/04/us/04talx.html`

[19] At the start of 2009, 6 out of 9 treatment states were at the federal minimum wage of $6.55, compared to only 1 out of 8 control states. This changed to 7 out of 9 treatment states and 6 out of 8 control states after the last federal increase to $7.25 in July 2009.

Because I do not observe firm-specific tax rates, I construct simulated measures of the effective payroll tax rate each employer would have faced at the maximum UI tax rate. The effective max rate in 2009 is calculated using each firm's actual payroll, while the simulated rates in 2010-13 are calculated fixing payroll at 2009 levels (the last year pre-policy change). Due to the difference in earnings, similar maximum UI tax costs at baseline translate into slightly higher effective tax rates in treatment states (2.58 versus 1.96). Nevertheless, we observe average simulated tax increases of 0.4, 1.2, and 1.4 percent of payroll from 2010 to 2012, while rates remain stable among control firms.

Finally, this table also illustrates differences in industrial composition across high- versus low-exposure firms. The three most common sectors among high-exposure firms are Construction, Manufacturing, and Administrative and Support Services. Construction is the most overrepresented, as nationally the industry only accounts for 10% of establishments, while among high-exposure firms construction employers make up 24% and among low-exposure firms they make up less than 3%. The administrative and support services sector is the second most overrespresented among high-exposure firms, while manufacturing is particularly overrepresented within treatment states. To account for these differences in industrial composition, I include fixed effects of 2-digit NAICS industry sector interacted with year-quarter in all regression specifications. The last few rows show that low-exposure firms are also more likely to be multi-establishment, are older on average, and have more educated workforces; these are the firm characteristics I will use to test for hetergenous responses in section 4.4.

3.3 First Stage

Although I cannot observe employer-specific tax rates, I provide evidence that increases to the statutory maximum UI tax translated into increases in actual taxes paid. Using industry-level data from the public-use Quarterly Census of Employment and Wages (QCEW), I estimate an event study regression using industries in the construction sector as a

proxy for high-exposure employers. Panel A of Figure 4 graphs statutory tax changes across treatment and control states, while Panel B plots event study estimates for the construction sector. State policy changes generally raised UI taxes for a three-year period, which led to an actual increase in effective UI tax rates precisely in the years following policy changes. I estimate effective UI tax rate increases of up to 0.5 percentage points in the three years post-policy change, relative to control states.[20] This average effect at the industry level underestimates the actual size of tax increases for high-exposure employers, since not all firms within construction would be facing the maximum tax rate. For comparison, Table 2 shows that high-exposure firms in treated states would experience a simulated rate increase of up to 1.4 percentage points if at the maximum tax rate.

In order to measure the first-stage effect on firm payroll, I construct simulated measures of each employer's effective payroll rate increase assuming they faced the maximum UI tax rate. Fixing payroll at 2009 levels, I simulate the UI tax increase a firm at the maximum tax rate would face in year t relative to the year 2009, and express this in terms of an effective tax rate, τ_{fst}. For example, in 2011 South Carolina raised its UI tax base from \$7,000 to \$10,000, and its maximum tax rate from 6.1% to 11.28%. A maximally exposed firm whose workers were previously earning \$30,000 annually would have faced an effective UI tax rate of \$427/\$30,000 = 1.42% under the old tax schedule. Under the new tax schedule, the firm faces an effective tax rate of \$1128/30,000 = 3.76%. This equates to a simulated increase of $\tau = 2.34\%$ in 2011. Meanwhile, τ_{fst} is defined to equal to zero in years 2008 and 2009.

Using τ_{fst} as my UI tax measure in regression estimates will provide an interpretable treatment effect in the absence of an observable first stage. Because τ_{fst} simulates the percentage change in labor costs that results from the state policy change, we can directly interpret the estimated magnitudes as labor demand elasticities under the assumption that each firm is subject to the maximum UI tax rate in year t. To the extent that the high-exposure sample is imputed and thus not all firms included in our analysis will be exposed

[20] Appendix Figure A.9 plots the same estimates at the quarterly rather than annual level.

to the maximum rate, this estimate produces a lower bound on the true labor demand elasticity. Additionally, if fixing payroll at 2009 levels overestimates the effective tax rates firms actually face (either due to wage growth or economic recovery over time), this will also tend to underestimate the true treatment effect.

4 Main Results

I first estimate triple-difference event study regressions to show the absence of pre-trends in treatment states relative to control states. I then estimate pooled triple-difference regressions to estimate a continuous treatment effect from UI tax increases.

4.1 Event Study

To estimate an event study, I group the nine treatment states into either 2010 tax changes or 2011 tax changes, and define $event = 0$ as the quarter prior to the tax change, either 2009:Q4 or 2010:Q4 respectively.[21] Event time always equals zero for control states. I then estimate the following event study regression specification:

$$\begin{aligned} Y_{fst} = \alpha_f + High_f * \mu_t + \sum_{k=-8, k\neq -1}^{13} \omega_k High_f * (event_s = k) \\ + \sum_{k=-8}^{13} \beta_k (event_s = k) + NAICS2_{ft} * \delta_t + \gamma minwage_{st} + \epsilon_{fst} \end{aligned} \tag{1}$$

Here f indexes firm, s indexes state, t indexes year-quarter, and k indexes the quarter relative to the policy change. Quarters more than two years before the policy change are assigned $event = -8$, and quarters more than three years after the policy change are assigned $event = 13$. To allow for potential anticipation effects, $k = -1$ is the omitted event quarter; thus the first treated quarter is labeled $k = 1$ and treatment effects are estimated relative to the outcome two quarters prior to the tax change (ie: if a state increased taxes in 2010,

[21] States with 2010 increases are Arkansas, Maine, Maryland, Tennessee, and West Virginia; States with 2011 increases are Illinois, Indiana, Oklahoma, and South Carolina.

the baseline would be 2009:Q3, and if a state increased taxes in 2011, the baseline would be 2010:Q3). Coefficients β_{-8} to β_{-2} (and ω_{-8} to ω_{-2}) test for differential pre-trends prior to the policy changes.

The inclusion of the $High_f$ indicator for high-exposure firms allows us to estimate triple-difference coefficients of interest ω_k, while the typical difference-in-differences coefficients β_k test for confounding conditions in treatment states relative to control states. Fixed effects are included for each firm, and fixed effects for year-quarter interacted with 2-digit NAICS industry codes (akin to industry sectors) control for industries such as construction being disproportionately impacted by the Great Recession. After the inclusion of high-exposure by year-quarter and industry by year-quarter fixed effects, any remaining threats to identification would have to operate through affecting high-exposure employers only in states that increased UI taxes.

Figure 5 plots estimates from Equation 1 using year-over-year employment growth as the outcome. The estimated coefficients show no differential pre-trend prior to the tax increases, and a sustained decline in employment growth afterwards. Highly exposed employers in treatment states experienced a 2 percentage point drop by the end of the first year, and a 5 percentage point drop after the second. Meanwhile, appendix Figure A.10 plots estimates for mean log earnings, and shows no differential impact on quarterly earnings at the firm level. To investigate this further, I estimate earnings effects at the worker-level in Section 5.

The growing employment effect over time could be driven by a few factors. First, average firm-specific tax rates did not peak until 2011-2012, as layoffs from the Great Recession took a few years to translate into higher tax rates. Thus, 2010 policy changes may not have become fully binding for firms until the second year after the policy change (Panel B of Figure 4 also shows the largest tax increase in year 2). Second, worker-level analysis in Section 5 finds partial incidence on earnings for low-earning workers in the very first quarter after the tax increase. If firms are able to shift some of the immediate tax burden onto workers through lower earnings, this would also lessen the initial impact on employment.

Finally, there could be employment hysteresis of the sort documented in Saez et al. (2021) from payroll tax cuts, and Yagan (2019) from the Great Recession. For example, UI payroll tax increases could result in a permanent decrease in labor demand if employers adjust through capital-labor substitution or outsourcing. Although I am unable to observe this, another possible explanation for the sustained drop in employment growth is substitution to contract workers, who would not be subject to UI taxation.[22]

As further evidence of causal identification, Figure A.11 plots difference-in-differences event study estimates by splitting the analysis sample into high-exposure and low-exposure subgroups. Panel A shows a similar treatment effect for highly exposed firms, while in Panel B we observe no impact on the placebo sample of low-exposure employers. In fact, the low-exposure event study suggests that in the absence of a policy change, highly exposed firms in treatment states might have fared slightly better than their counterparts in control states.

4.2 Pooled Triple-Difference

After showing the absence of differential pre-trends, I next estimate a pooled triple-difference regression to allow for a continuous treatment measure, τ_{fst}, defined as the simulated increase in the firm's effective UI tax rate.

$$\begin{aligned} Y_{fst} = \alpha_f + High_f * \mu_t + \omega High_f * \tau_{fst} + \beta \tau_{fst} \\ + NAICS2_{ft} * \delta_t + \gamma minwage_{st} + \epsilon_{fst} \end{aligned} \tag{2}$$

Here f indexes firm, s indexes state, and t indexes year-quarter. ω is the triple-difference coefficient of interest, while β represents the difference-in-difference estimate. Consistent with the event study specification, I include a control for the state minimum wage, as well as firm fixed effects, high-exposure by year-quarter fixed effects, and 2-digit NAICS industry by year-quarter fixed effects. This specification estimates $\hat{\omega}$ = -2.24 (significant at the 1 percent level), implying that a one percent increase in the effective tax rate lowers year-over-year

[22]For example, Kugler et al. (2017) found that payroll tax cuts in Colombia increased rates of formal employment among affected workers.

employment growth by 2.24 percentage points. This represents the causal impact of a UI tax increase, under the assumption that any unobservables that specifically affect the labor demand of high-exposure employers (compared to low-exposure employers) are uncorrelated with the size of legislated UI tax increases. However, the estimated treatment effect is a lower bound on the true labor demand elasticity due to measurement error from the imputation of firms' UI tax exposure.

I also interact τ_{fst} with quarterly indicators to estimate four triple-difference coefficients of interest ω_1 to ω_4, which tests for differential responses by calendar quarter. This is of interest because firm-level UI tax burdens are largest in Q1, and worker-level tax burdens for new hires are largest in Q4.

$$Y_{fst} = \alpha_f + High_f * \mu_t + \sum_{q=1}^{4} \omega_q * High_f * \tau_{fst} + \sum_{q=1}^{4} \beta_q * \tau_{fst} + NAICS2_{ft} * \delta_t + \gamma minwage_{st} + \epsilon_{fst} \tag{3}$$

Table 3 reports estimates from Equation 3 for three different employment outcomes as well as log earnings. The regressions are unweighted, since the analysis sample is already restricted to employers between 20 and 500 workers at baseline, although weighting by employment has little impact on the estimated magnitudes (weighted regressions are reported in Appendix Table A.1). In Column 1, the positive coefficients in the first four rows indicate that in the absence of a policy change, firms in treated states would have fared slightly better than their counterparts in control states. However, the triple-difference coefficients in the next four rows show that a one percentage point increase in the effective tax rate actually lowered year-over-year employment growth by 2.53 percentage points by the end of the year. To decompose this employment change into hires and separations, I define the quarterly new hire share as the number of new hires divided by employment, and the quarterly separation share as the number of separations divided by employment. We observe a negative impact on hiring, with the largest reductions at the end of the calendar year. Meanwhile, Column

3 finds no impact on separations, and Column 4 also estimates no impact to mean earnings.

This pattern of negative employment growth and hiring is consistent with the mechanism of greater UI tax burdens. At the start of the calendar year all earnings up to the tax base are eligible for UI taxes, and thus create a large payroll tax burden for the employer. Each additional worker hired incurs an additional UI tax burden, whereas additional separations do not incur a tax saving and could even push future costs higher if they claim UI benefits. Towards the end of the calendar year, most if not all existing workers have exceeded the UI tax base and no longer cost their employer any additional payroll tax. However, the earnings of new hires still incur taxes up to the same tax base, resulting in a much higher effective tax rate for new hires; this could explain the observed reluctance to hire in the last two quarters of the year. Appendix Figure A.12 plots event study estimates of the new hire share, which shows a similar pattern. Meanwhile Figure A.13 shows that the effect on new hires is only present within the high-exposure subgroup and not the low-exposure subgroup.

Additionally, I create an alternative measure of the state UI tax change, $Tax\Delta_{st}$, which measures in dollar terms the difference between the maximum UI tax rate in year t relative to year 2009. In the prior example for South Carolina, $Tax\Delta_{st}$ would be calculated as \$1128 - \$427 = \$701. Substituting this measure for τ_{fst} in the previous equation gives an alternative regression specification to test the sensitivity to how UI tax changes are measured. Table 4 reports triple-difference estimates using this alternative treatment measure, and results in extremely similar patterns of negative employment growth and hiring, coupled with no response on the separations and earnings margins.

To calculate an implied labor demand elasticity for comparison, we can divide the triple-difference drop in Q4 employment growth of -0.74 by 0.31, the share of average annual earnings a \$100 tax increase is equivalent to. This equates to an end-of-year labor demand elasticity of -2.4, which is quite similar to the estimated Q4 magnitude of -2.53 from Column 1 of Table 3. Results are also robust to the inclusion of state-by-year-quarter fixed effects,

reported in Table A.2.[23]

4.3 Heterogeneity by Worker and Employer Characteristics

Given the estimated negative effect on employment growth, I next test for treatment effect heterogeneity for specific subgroups of workers. First, I study the impact on the hiring rate of low-earning workers, who are defined to be those who earned less than $3000 in the quarter after being hired (inflation adjusted to 2015 dollars).[24] Next I calculate year-over-year employment growth rates for two subgroups of interest: workers younger than 25 years of age, and those with only a high school education or less (conditional on being 25 or older). If firms did not employ these types of workers in either the current or previous calendar quarter, growth rates are automatically defined to equal zero.

Table 5 reports estimates for these three outcomes using the $Tax\Delta_{st}$ measure of tax increases. Column 1 shows only a marginally significant effect for low-earning new hires in Q1; when benchmarked to the sample mean of 5.3%, the magnitude implies a similar effect as that of the overall share of new hires from the previous table (0.16pp relative to a 10.6% mean). Recall the previous triple-difference estimate of a $100 increase in maximum UI taxes lowering employment growth in Q4 by 0.74 percentage points (Column 1 of Table 4). For young workers, Column 2 estimates that the impact is an even larger decline of 0.90 percentage points for each $100 increase. Growth rates for low-educated workers, on the other hand, are more similar to the overall effect, with estimated coefficients in Column 3 that are similar in magnitude to the estimates for overall employment growth (-0.75pp by Q4 for each $100 increase). This suggests that young workers who are just entering the workforce are the most impacted by falls in labor demand due to UI tax increases.

To test for heterogeneous responses by different types of employers, I classify employers based on their characteristics prior to the policy changes (as of 2009:Q3). I focus on three

[23] Additionally, difference-in-difference estimates for the high-exposure subgroup are reported in Appendix Table A.3.

[24] The $3000 threshold was chosen to coincide with the federal poverty level for a single person household ($11,770 in 2015).

observable characteristics: multi-establishment status, firm age, and education level of the firm's workforce. The first is also a potential proxy for whether the employer would face cash or financing constraints, as a surprise increase in UI taxes could restrict available cash flow.

I augment Equation 2 to estimate a regression specification that interacts the triple-difference coefficient with an indicator for having a given employer characteristic. For example, the following equation estimates heterogeneous effects by single-establishment status $Single_f$:

$$\begin{aligned} Y_{fst} = \alpha_f + High_f * \mu_t + \rho Single_f * High_f * \tau_{fst} + \omega High_f * \tau_{fst} \\ + \beta \tau_{fst} + NAICS2_{ft} * \delta_t + \gamma minwage_{st} + \epsilon_{fst} \end{aligned} \tag{4}$$

Analogous regressions are also estimated using an indicator for firm age greater than 20 (Old_f), and an indicator for more than 40 percent of their workforce having no college education ($NoCollege_f$).[25]

The interaction coefficient estimates are plotted in Figure 6, with the full table of estimates in Table 6. In terms of employment growth, single-establishment and older firms are the most responsive to UI tax increases; the estimated interaction terms imply that they are roughly twice as responsive as multi-establishment firms and young firms. Meanwhile, the education level of a firm's workforce has no impact on its employment response. On the new hire margin, single-establishment firms are also more responsive than multi-establishment firms, with a one percentage point increase in payroll lowering the share of new hires by 0.33 percentage points (and no statistically significant main effect for multi-establishment firms). This aligns with the argument that firms with less access to internal resources are the most likely to decrease hiring in response to UI tax increases. Finally, there is no evidence of differential impacts on the separations margin.

[25]These cutoffs were chosen such that at baseline, roughly half of the sample of high-exposure firms were on each side. Recall that all firms in the analysis sample are already age 3 or older by 2009, as I used employment history from 2006 to 2009:Q2 to impute UI tax exposure.

5 Worker-level Tax Incidence

To study payroll tax incidence on earnings with greater precision, this section estimates regressions at the worker rather than firm level. In this context, incidence on workers includes anything that affects earnings rather than employment; if firms do not adjust wages but lower the number of hours, this will also be captured in the earnings estimates as worker incidence. Due to differences in worker tenure and earnings among high- and low-exposure firms, I restrict my analysis to difference-in-differences event studies of only workers at high-exposure firms.

In order to compare earnings before and after the tax increases, I limit my analysis to existing stable workers – employed in both 2008 and 2009, with annual earnings between \$5,000 and \$100,000 in 2009 – and estimate the impact of UI tax increases on their subsequent earnings at the same firm. I estimate a worker-level difference-in-differences event study regression, where event time is defined in the same way as in Equation 1. To incorporate a more continuous measure of treatment, I define $\bar{\tau}_f$ to equal the average of τ_{fst} taken over the post-period, and interact this measure with the event time indicators.

$$Log(Earn)_{ifst} = \mu_i + \alpha_f + \sum_{k=-8, k\neq -1}^{8} \beta_k \bar{\tau}_f (event_s = k) + \delta_t + \rho log(earn_{ifs,t-4}) + \gamma minwage_{st} + \epsilon_{ifst} \quad (5)$$

Here i indexes worker, f indexes firm, s indexes state, t indexes year-quarter, and k indexes the quarter relative to the policy change. Quarters more than two years before the policy change are assigned $event = -8$, and quarters more than two years after the policy change are assigned $event = 9$. Fixed effects are included for each worker, firm, state-quarter, and year-quarter, and also included are controls for the minimum wage and the previous year's quarterly earnings. The analysis sample only includes incumbent workers, and they are included only for the quarters they are employed at the firm. I also require firms to have

at least 10 incumbent workers to be included in the sample. This results in a sample of 1.4 million workers at 27,500 firms (over 90% of the high-exposure firms in the main analysis sample).

Figure 7 plots the estimates of the event time interactions with $\bar{\tau}_f$. All estimated coefficients are statistically insignificant and close to zero, implying an absence of earnings effects for incumbent workers. Appendix Figure A.14 also estimates a weighted version weighting by the inverse of firm employment at baseline, which serves to upweight workers in small firms, but similarly fails to find an earnings effect. This is consistent with the firm-level evidence of an absence of tax incidence on mean earnings.

5.1 Low-Earning Workers

Given the absence of tax incidence for the average worker, I next focus on a subgroup of low-earning workers. Due to low UI tax bases, low-wage workers face higher effective tax rates and thus bear a larger relative tax burden than their higher-earning counterparts. They may also have less bargaining power and are more likely to turn over or work part-time.

I restrict this subsample to incumbent workers who have annual earnings between $5000 and $24,000 in 2009.[26] To ensure these workers make up a non-negligible share of the firms' employment, I also drop any firms that did not employ 10 or more of these low-wage workers as of 2009:Q3. This subsample includes approximately 549,000 low-wage workers, employed by 13,000 employers (44% of the high-exposure firms in the main analysis sample).

Instead of using $\bar{\tau}_f$, which is sensitive to low earnings levels, I now measure tax increases using $Tax\Delta_s$, which is defined for treatment states to equal the average maximum UI tax increase in dollars between 2009 and the post-period (in hundreds). The main outcome of interest, $EarnGrowth_{ifst}$ is defined as the year-over-year earnings growth relative to the same quarter in the previous year. Thus inclusion in the sample requires the worker to have

[26]Because the LEHD reports total earnings and not wage rates, this definition may pick up high-wage workers who work very few hours, although this should be fairly uncommon. Additionally, since UI taxes are based on earnings and not wages, part-time high-wage workers are equally costly to an employer from a payroll tax standpoint.

been employed at the firm a year earlier, limiting the analysis to a more stable subset of workers.

Figure 8 plots the estimates of the event time interactions with $Tax\Delta_s$; thus the magnitude of each estimate reflects the effect of a $100 UI tax increase. In the first quarter after the tax increase ($event = 1$), each $100 UI tax increase lowered earnings growth by 0.63 percentage points for continuing workers in treated firms, relative to their counterparts in control firms. This negative impact is small and precisely estimated, but does not last beyond the first year. And because this subgroup is likely to be working hourly and/or part-time jobs, the drop in earnings could reflect a reduction an hours rather than a reduction in wages. Therefore, even amongst a subgroup that bears the largest effective tax burden, tax increases result in less than full tax incidence on the earnings of existing workers. In contrast, estimating this regression on alternate subgroups of medium- or high-earning workers results in no perceivable treatment effect.

6 Conclusion

The employer-specific and time-varying aspects of UI payroll taxation provide researchers with extensive policy variation that can be used to understand how firms respond to unexpected payroll tax increases. Furthermore, the experience rating of UI taxes based on layoff history causes any UI payroll tax increase to target a particularly vulnerable subset of the firm population.

Comparing highly exposed firms across treatment and control states, I estimate that unexpected state tax increases resulted in significant negative employment responses, especially for single-establishment employers with less access to financial resources. With little ability to pass tax increases on to workers, employers lowered hiring rates and employment growth, and this effect grew over time. Using a pooled triple-difference regression, I estimate a lower bound on the labor demand elasticity of 2.4. This elasticity estimate falls within the

range of elasticities from the labor demand literature. Ku et al. (2020), estimates a labor demand elasticity of 3.6 using place-based payroll tax increases in Norway, while Benzarti and Harju (2021*a*) estimates elasticities of 2.9 to 4.16 in response to firm-specific increases in Finland. And in the U.S. setting, Johnston (2021) estimates a UI tax labor demand elasticity of 4 while Guo (2023) estimates an elasticity of 1.1. Given the frequent proposals to cut employer payroll taxes as a way stimulate employment during recessions, my findings suggest that providing highly exposed firms with UI payroll tax relief could be an even more effective policy response.

The tax incidence of unemployment insurance payroll taxes has been an understudied topic in the payroll taxation literature. The experience rating of employer-specific tax rates creates the benefit of reducing avoidable layoffs (particularly temporary layoffs) and the associated externalities imposed upon the UI system. However, the cyclical pattern in which tax rates are highest in years following economic recessions also slows hiring and employment growth during the recovery. More work is needed to understand the unintended consequences and inherent tradeoffs of UI's current design in the United States.

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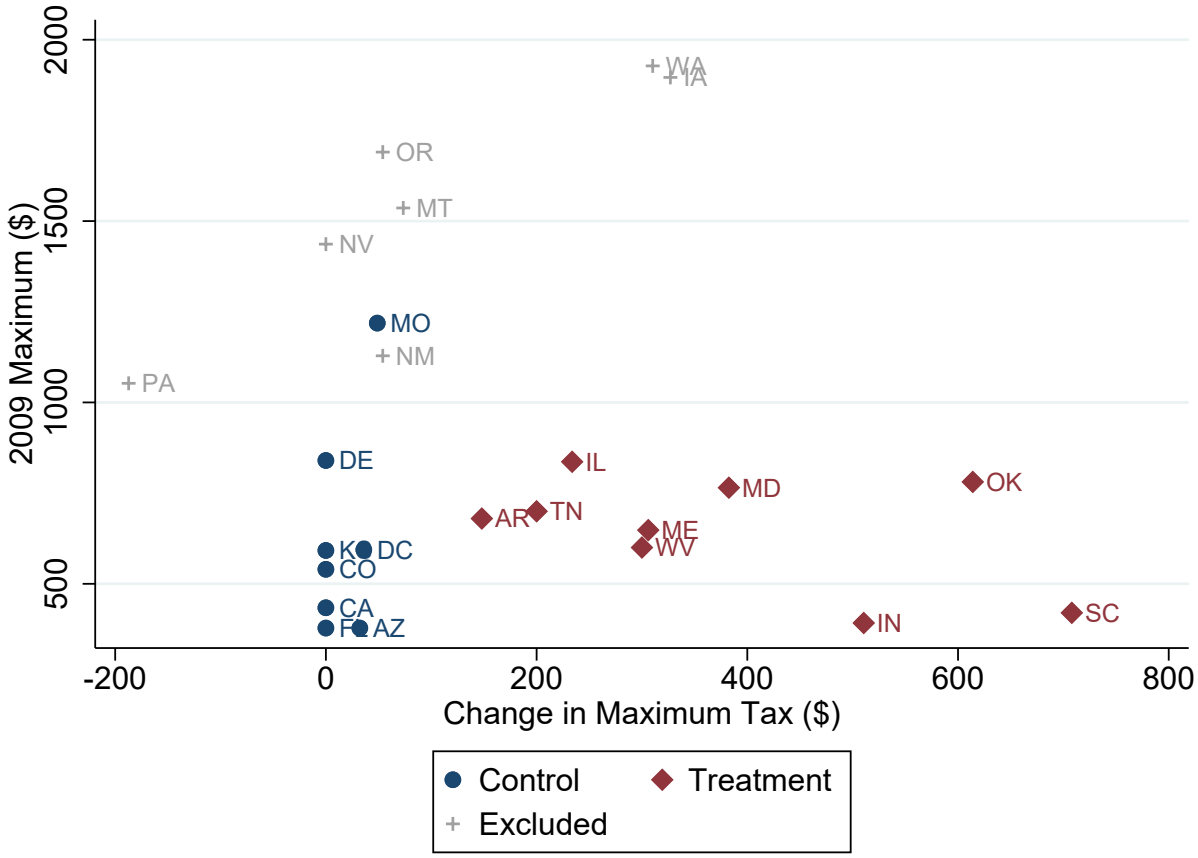
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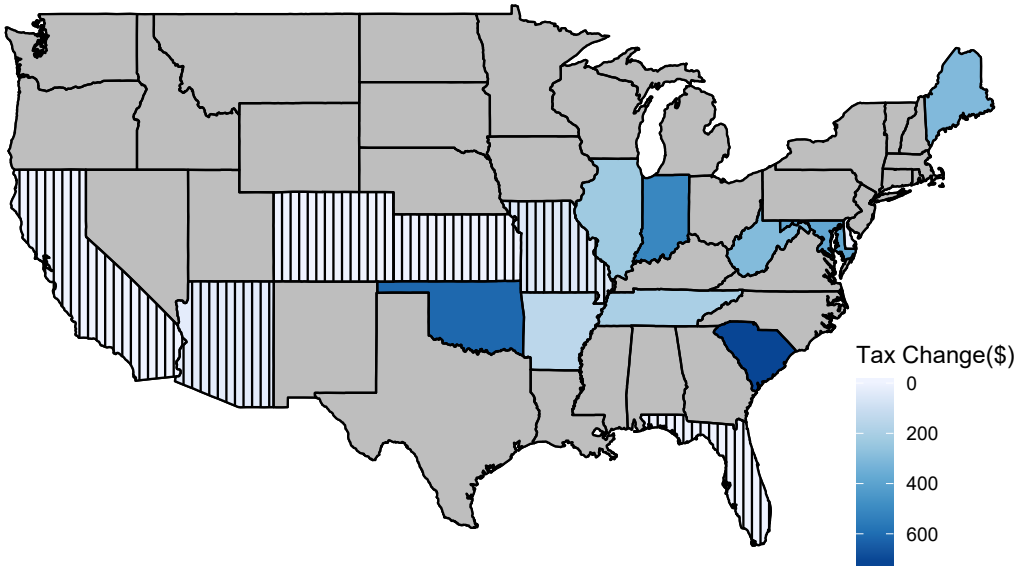
FIGURES

Figure 1: Maximum UI Tax Increase from 2009 to 2011



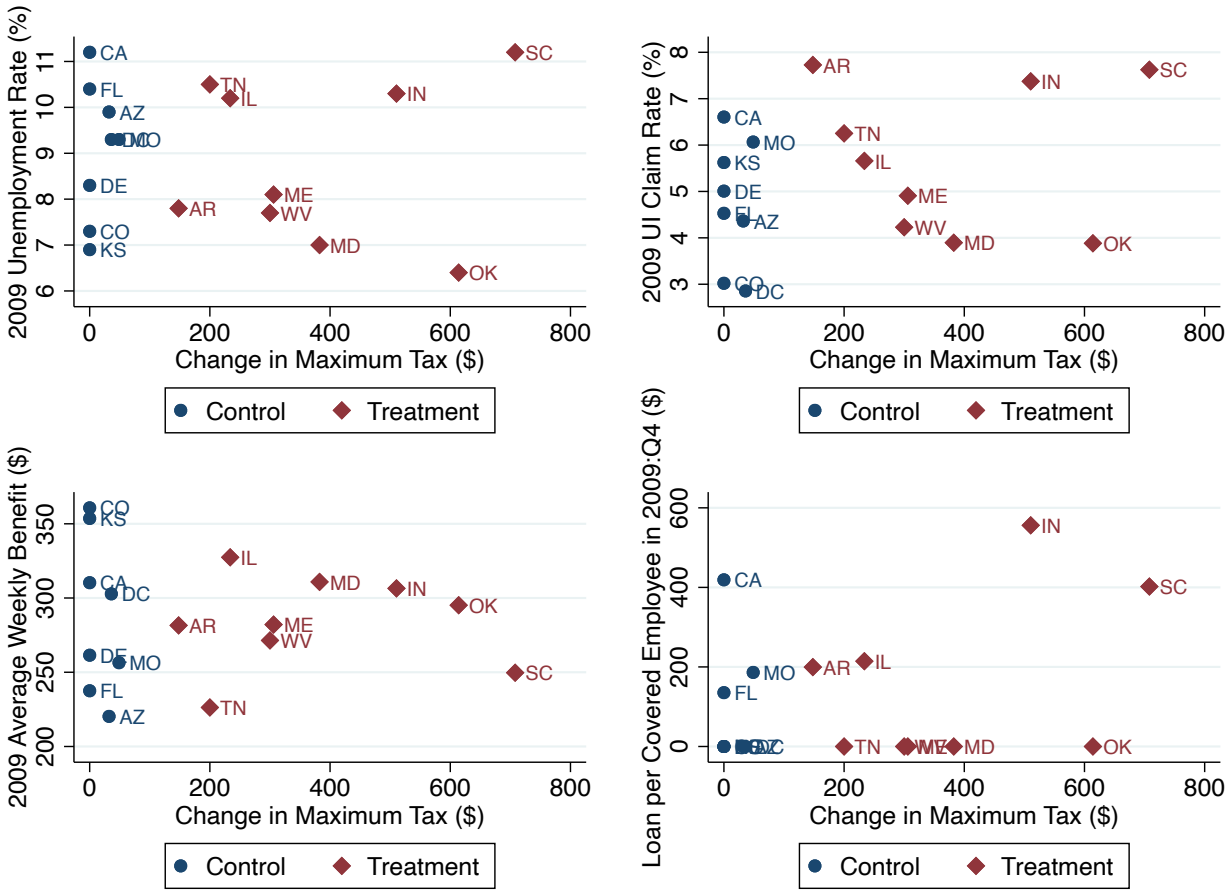
Source: *Significant Measures of State UI Tax Systems*. This figure plots each state's change in maximum tax from 2009 to 2011 against its 2009 maximum tax level. The maximum tax is calculated as taxable wage base multiplied by the maximum tax rate. Six states are excluded because their tax schedules are automatically indexed to inflation, and Pennsylvania is excluded because it experienced a tax decrease. Nine states with changes greater than $100 are categorized as Treatment states, while the remaining eight states are Control states.

Figure 2: UI Tax Policy Variation



Source: *Significant Measures of State UI Tax Systems*. This map plots each state's change in maximum tax from 2009 to 2011. The nine treatment states are shaded in blue corresponding to the tax change they experienced, and the eight control states are shaded with vertical lines (and also include Delaware and District of Columbia). States in grey are excluded.

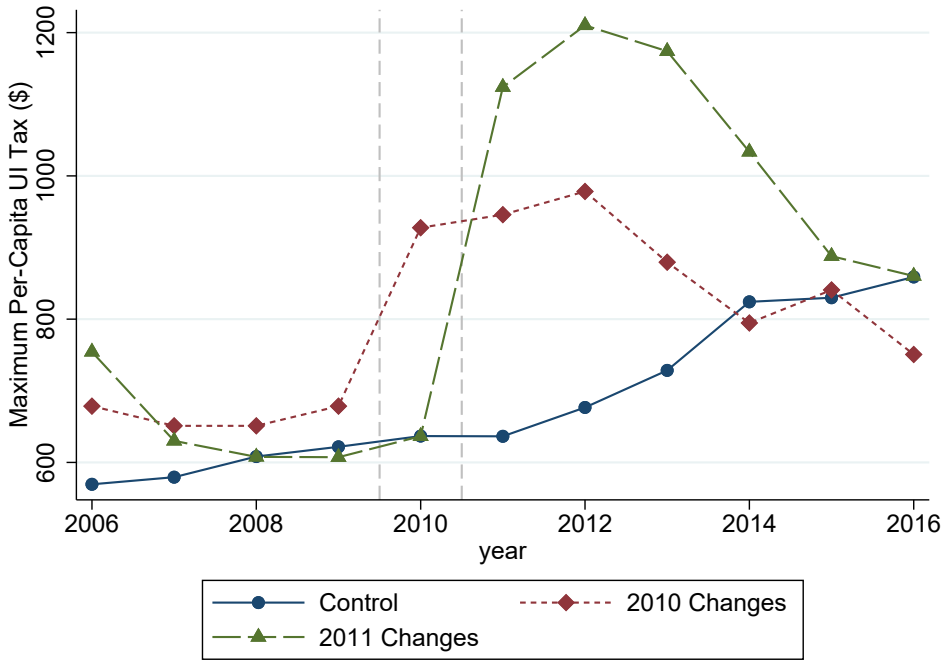
Figure 3: Exogeneity of State UI Tax Increases



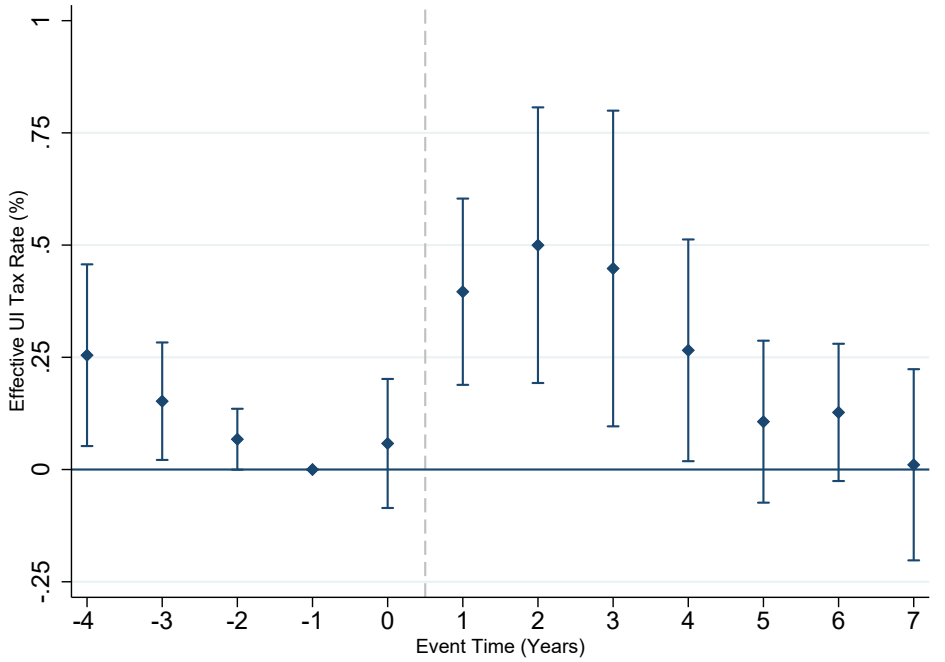
This figure plots the change in the maximum tax from 2009 to 2011 against four different measures of state economic conditions in 2009: the annual unemployment rate, the share of initial UI claims divided by covered employment, the average weekly UI benefit, and the UI trust fund loans per covered worker. The first measure is sourced from the Bureau of Labor Statistics' *Local Area Unemployment Statistics*, while the last three measures are sourced from the Department of Labor's quarterly *UI Data Summary*.

Figure 4: State UI Tax Increases Lead to Actual Increases in UI Costs

A: Statutory Maximum UI Taxes (2006-2016)

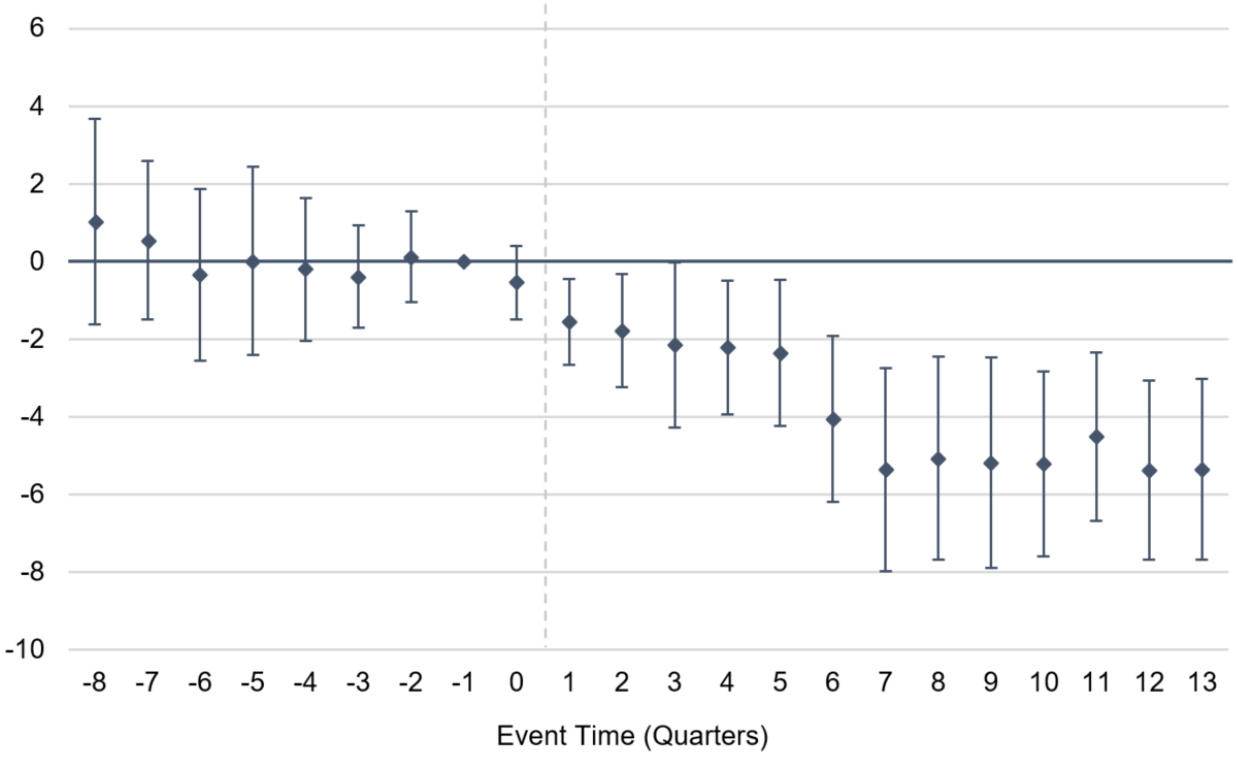


B: Event Study of UI Tax Rates (2006-2016)



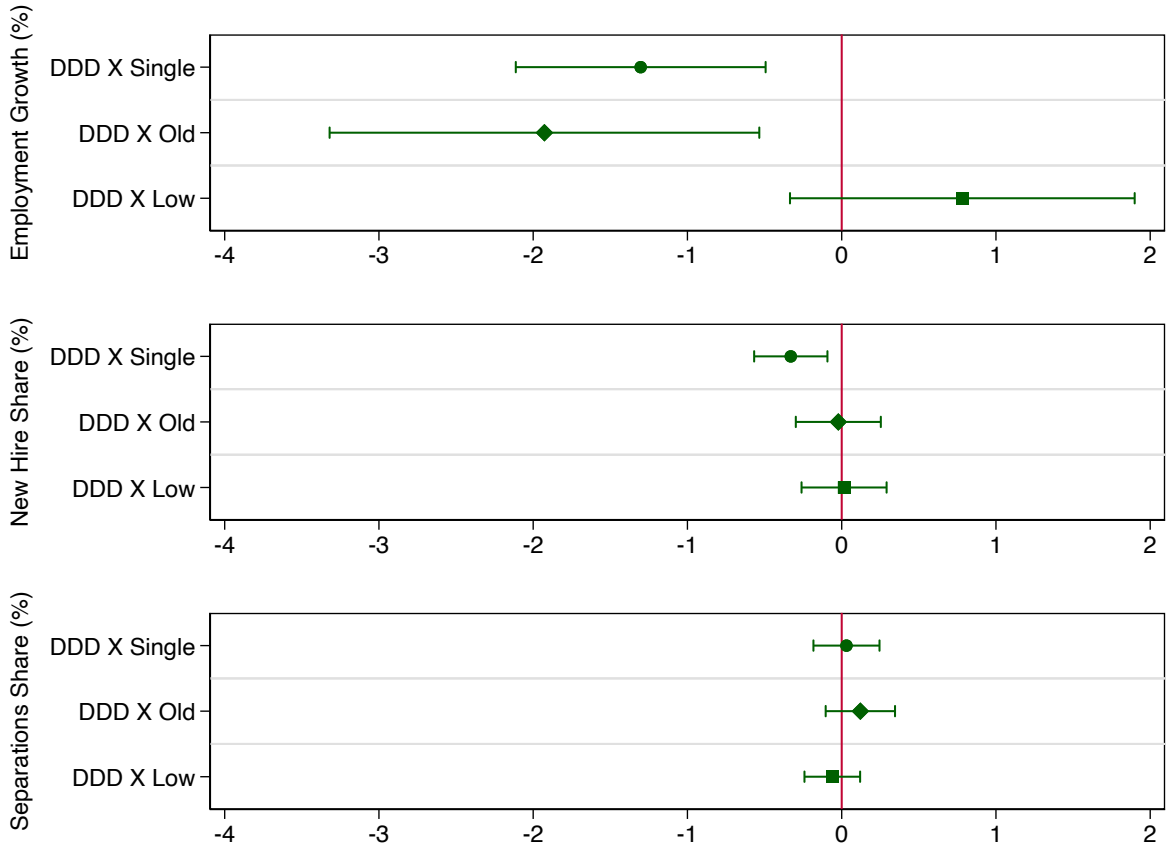
Panel A plots the employment-weighted average statutory maximum UI tax (maximum rate * tax base) across control states, states with 2010 policy changes, and states with 2011 policy changes. Panel B estimates an event study of effective UI tax rates, with state-industry and year fixed effects, and weighted by industry employment (N = 1,846). Sample limited to ten 4-digit construction industries (cells too small to meet disclosure requirements are missing), and effective tax rates are calculated by dividing the sum of annual UI contributions by annual payroll. Error bars denote 95% CI for standard errors clustered at state level.

Figure 5: Triple-Difference Event Study Estimates of Employment Growth (2008-2013)



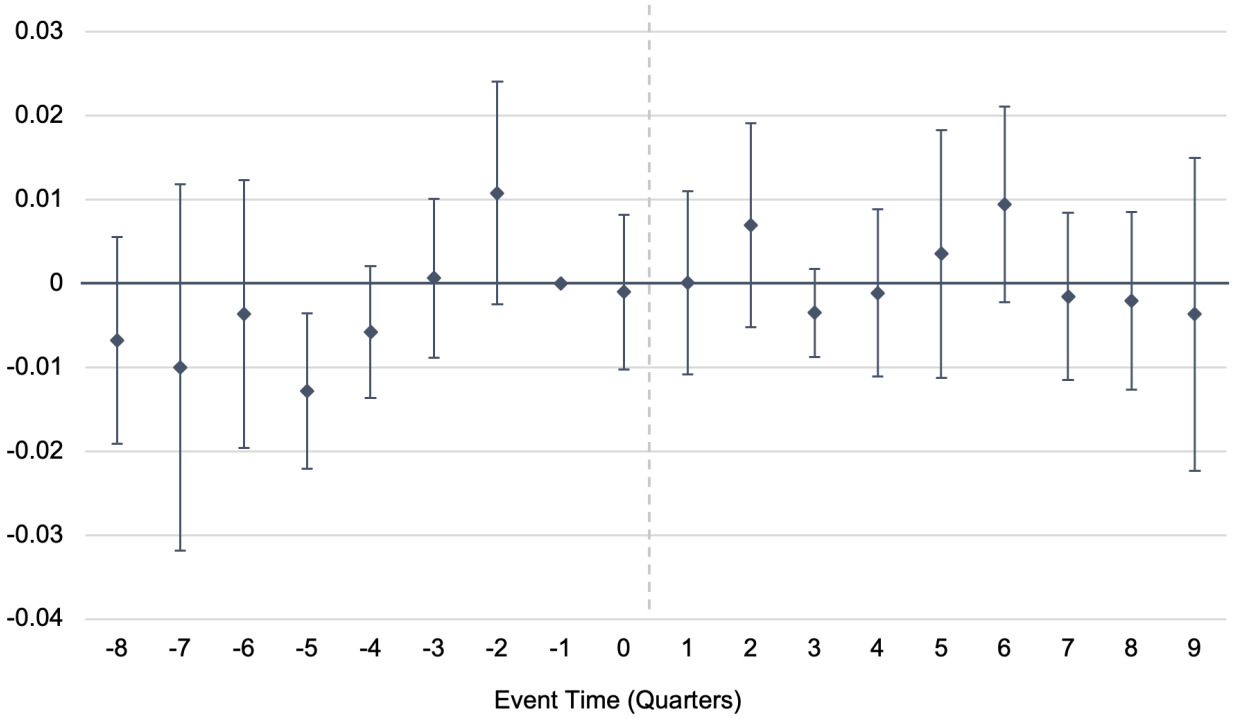
N = 1,109,000. Figure plots the estimates from treatment dummies interacted with event time and high-exposure status. Regression also includes treatment dummies interacted with event time, a control for state minimum wage, and fixed effects for firm, NAICS2-year-quarter, and high-year-quarter. Error bars denote 95% CI for robust standard errors clustered at state level.

Figure 6: Heterogeneity by Firm Characteristics



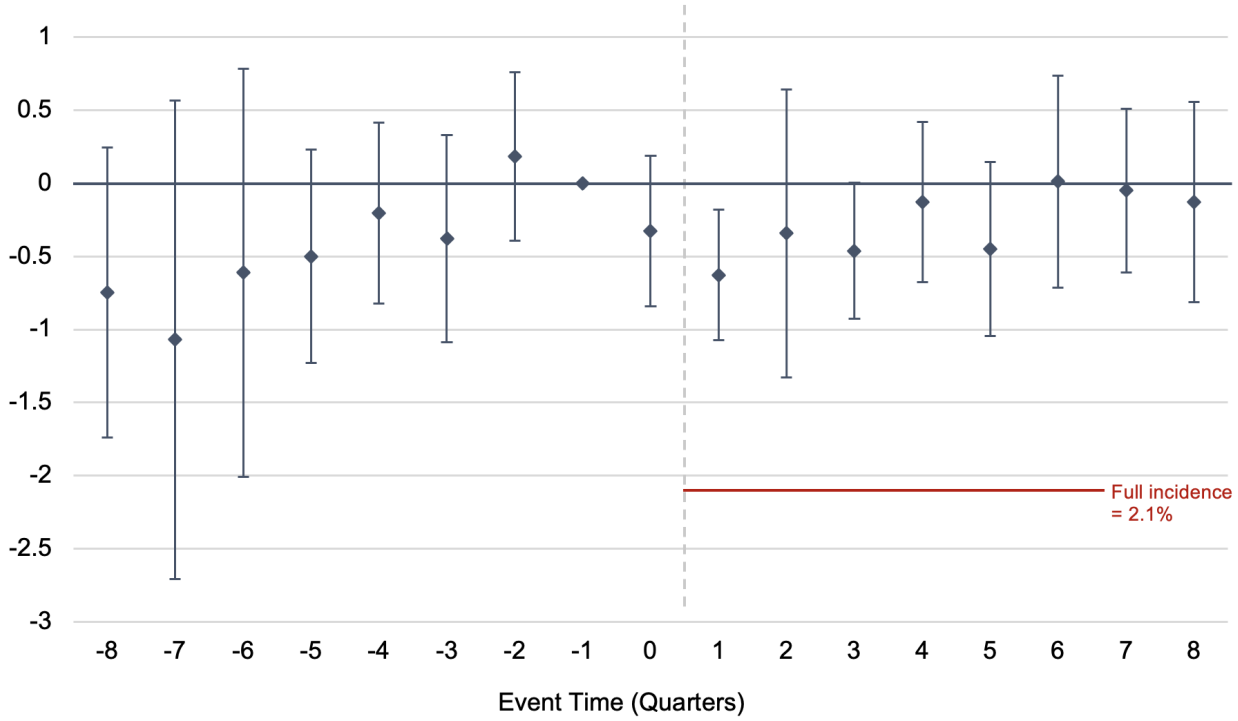
N = 1,109,000. Figure plots estimates from nine separate regressions of the triple-difference estimate, $\tau \times$ High, interacted with the respective firm characteristic. τ measures the firm's simulated effective rate increase from 2009 to the current year. High is an indicator for high-exposure employers. Single is an indicator for single-establishment firm (65.8% of sample). Old is an indicator for firm age older than 20 (56.9% of sample). Low is an indicator for more than 40 percent of firm's workforce having no college education (40.3% of sample). Regressions include firm, NAICS2-year-quarter, and high-year-quarter fixed effects. Error bars denote 95% CI for robust standard errors clustered at state level.

Figure 7: Log Earnings - Worker Level (2008-2012)



N = 17,190,000. Includes 1,401,000 unique workers across 27,500 unique employers. Figure plots the estimates from $\overline{\tau}_f$ interacted with event time. Regression also includes a control for state minimum wage, lagged log earnings, and firm, year-quarter, and state-by-calendar quarter fixed effects. Mean log quarterly earnings equals 9.08. Error bars denote 95% CI for robust standard errors clustered at state level.

Figure 8: Earnings Growth for Low-Earning Workers (2008-2012)



N = 4,954,000. Includes 549,000 unique workers across 13,000 unique employers. Figure plots the estimates from $Tax\Delta_s$ interacted with event time. Regression also includes a control for state minimum wage, lagged log earnings, and firm, year-quarter, and state-by-calendar quarter fixed effects. Mean log quarterly earnings equals 8.47. Error bars denote 95% CI for robust standard errors clustered at state level.

TABLES

Table 1: Mean Quarterly UI Tax Contributions for Construction Sector (2011)

	Control - $7,850 base		Treatment - $11,200 base	
	Contribution per Worker	Effective Tax Rate	Contribution per Worker	Effective Tax Rate
Q1	$380	3.3%	$452	4.2%
Q2	$121	1.0%	$195	1.8%
Q3	$83	0.66%	$126	1.1%
Q4	$58	0.45%	$87	0.7%

Source: *Quarterly Census of Employment and Wages*. Contributions per worker calculated as quarterly contributions divided by monthly employment. Effective tax rate calculated by dividing total quarterly contributions by total quarterly wages. Control states include: AZ, CA, CO, DE, DC, FL, KS, and MO. Treatment states include: AR, IL, IN, ME, MD, OK, SC, TN, and WV. Tax base calculated as the employment-weighted mean across treatment and control states.

Table 2: Summary Statistics (2009:Q3)

	High-Exposure Subgroup				Low-Exposure Subgroup			
	Control		Treatment		Control		Treatment	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Employment	66.74	72.28	67.29	70	76.07	85.18	85.06	94.22
Layoff Share 2006-07	56.68	16.89	56.68	16.79	8.38	4.12	8.24	4.05
Layoff Share 2008-09	50.37	14.73	50.65	14.93	5.58	2.95	5.50	2.91
Average Annual Earnings	36,900	27,990	32,350	23,340	59,280	60,910	52,520	57,740
Effective Max Rate 2009	1.963	1.327	2.58	1.233	1.519	1.333	1.892	1.173
Simulated Max Rate 2010	1.991	1.354	3.025	1.548	1.538	1.357	2.186	1.416
Simulated Max Rate 2011	1.99	1.354	3.768	1.565	1.538	1.357	2.719	1.564
Simulated Max Rate 2012	2.1	1.378	3.978	1.767	1.611	1.393	2.923	1.712
Simulated Max Rate 2013	2.195	1.464	3.702	1.679	1.657	1.438	2.743	1.604
Employment Growth	-19.1	23.52	-17.93	23.67	9.432	26.74	6.628	22.8
New Hire Share	11.83	11.24	12.72	11.74	7.737	11.79	7.159	10.28
Separations Share	15.79	12.59	16.69	13.39	8.177	12.33	7.895	11.24
Construction Share	24.5		22.82		2.22		2.69	
Manufacturing Share	12.65		18.28		12.39		11.0	
Admin/Support Share	12.13		13.02		4.59		3.92	
Single-Establishment	70.41		66.99		62.41		60.59	
Firm Age 20+	47.47		54.17		61.88		69.71	
Less Educated	48.07		57.87		23.05		24.75	
N	18,000		11,500		9,900		10,000	

Employer-level observations from 3rd quarter of 2009. Restricted to high exposure employers, with employment between 20 and 500 workers in 2009:Q3. Control states include: AZ, CA, CO, DE, DC, FL, KS, and MO. Treatment states include: AR, IL, IN, ME, MD, OK, SC, TN, and WV. Effective max rate 2009 is calculated as the firm's effective payroll tax rate if they paid the maximum UI tax rate. Simulated max rate 2010-2013 calculates the firm's effective payroll tax rate under the maximum UI tax rate in 2010-2013 respectively, but using payroll from 2009. Less Educated is an indicator for having a workforce where more than 40 percent have no college education.

Table 3: Pooled Triple Difference Estimates (2008–2013)

	Employment Growth	New Hire Share	Separations Share	Log Earnings
	(1)	(2)	(3)	(4)
$\tau \times$ Q1	1.021**	-0.386***	-0.524**	0.00116
	(0.382)	(0.0897)	(0.227)	(0.00518)
$\tau \times$ Q2	0.991**	0.286	-0.0719	0.0223**
	(0.398)	(0.237)	(0.281)	(0.00886)
$\tau \times$ Q3	0.906*	0.106	0.0655	0.00943
	(0.486)	(0.131)	(0.276)	(0.0131)
$\tau \times$ Q4	0.958*	-0.0182	-0.261**	-0.00315
	(0.501)	(0.0926)	(0.107)	(0.00629)
$\tau \times$ Q1 $\times$ High	-1.875***	-0.305*	0.144*	-0.00458
	(0.565)	(0.158)	(0.0808)	(0.00489)
$\tau \times$ Q2 $\times$ High	-2.292***	0.370	-0.141	-0.00632
	(0.613)	(0.355)	(0.199)	(0.00874)
$\tau \times$ Q3 $\times$ High	-2.281***	-0.545***	0.235	0.00558
	(0.701)	(0.175)	(0.309)	(0.0133)
$\tau \times$ Q4 $\times$ High	-2.527***	-0.689**	0.412	0.0109*
	(0.756)	(0.248)	(0.421)	(0.00584)
Minimum Wage	-0.261	-0.122	0.00347	0.0106***
	(0.324)	(0.116)	(0.138)	(0.00278)
R^2	0.203	0.476	0.483	0.901
Mean of Dep Variable	-1.373	10.61	11.28	9.323
N	1109000	1109000	1109000	1109000

Dependent variables are multipled by 100 (and thus denoted in percentage points). τ measures the firm's simulated effective rate increase from 2009 to the current year. High is an indicator for high-exposure employers. Regressions include firm, NAICS2-year-quarter, and high-year-quarter fixed effects. The necessary DDD interactions are either absorbed by the firm fixed effects or the high-year-quarter fixed effects. Robust standard errors clustered at state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Triple Difference Estimates Using Dollar Increase (2008–2013)

		Emp Growth	New Hire Share	Sep Share	Log Earnings
		(1)	(2)	(3)	(4)
TaxΔ*Q1	($100's)	0.330**	-0.0286*	-0.0385	-0.000610
		(0.116)	(0.0143)	(0.0373)	(0.00147)
TaxΔ*Q2	($100's)	0.303**	0.0739	0.0305	0.00313
		(0.127)	(0.0660)	(0.0624)	(0.00274)
TaxΔ*Q3	($100's)	0.315**	0.0382	-0.0480	-0.000553
		(0.135)	(0.0289)	(0.0485)	(0.00283)
TaxΔ*Q4	($100's)	0.284*	0.0380	-0.0719***	0.00291**
		(0.140)	(0.0231)	(0.0240)	(0.00128)
TaxΔ*Q1 $\times$ High		-0.503**	-0.155**	-0.0544	-0.000246
		(0.190)	(0.0654)	(0.0508)	(0.00172)
TaxΔ*Q2 $\times$ High		-0.622***	0.155	-0.0995	0.00112
		(0.212)	(0.156)	(0.0574)	(0.00261)
TaxΔ*Q3 $\times$ High		-0.664***	-0.151**	0.110	0.00433
		(0.222)	(0.0596)	(0.0797)	(0.00326)
TaxΔ*Q4 $\times$ High		-0.743***	-0.220**	0.135	0.00223
		(0.234)	(0.101)	(0.164)	(0.00172)
Minimum Wage		-0.383	-0.170	-0.00830	0.0108***
		(0.336)	(0.118)	(0.144)	(0.00271)
R^2		0.203	0.476	0.482	0.901
Mean of Dep Variable		-1.373	10.61	11.28	9.323
N		1109000	1109000	1109000	1109000

Dependent variables are multipled by 100 (and thus denoted in percentage points). TaxΔ measures the dollar change in the state maximum UI tax from 2009 to the current year. High is an indicator for high-exposure employers. Regressions include firm, NAICS2-year-quarter, and high-year-quarter fixed effects. The necessary DDD interactions are either absorbed by the firm fixed effects or the high-year-quarter fixed effects. Robust standard errors clustered at state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Heterogeneity by Worker Characteristics (2008–2013)

		Emp Share: Low-Earning Hires	Emp Growth: Age under 25	Emp Growth: No College
		(1)	(2)	(3)
TaxΔ*Q1	($100's)	-0.0295**	0.379	0.371**
		(0.0126)	(0.228)	(0.135)
TaxΔ*Q2	($100's)	0.0414	0.295	0.309*
		(0.0468)	(0.287)	(0.154)
TaxΔ*Q3	($100's)	-0.0524*	0.208	0.357**
		(0.0287)	(0.242)	(0.149)
TaxΔ*Q4	($100's)	0.00105	0.222	0.299**
		(0.0168)	(0.265)	(0.139)
TaxΔ*Q1 × High		-0.0796*	-0.618*	-0.525**
		(0.0397)	(0.293)	(0.233)
TaxΔ*Q2 × High		0.0490	-0.659**	-0.617**
		(0.0534)	(0.290)	(0.274)
TaxΔ*Q3 × High		-0.0212	-0.750**	-0.705**
		(0.0455)	(0.326)	(0.253)
TaxΔ*Q4 × High		-0.0890	-0.904**	-0.748***
		(0.0517)	(0.387)	(0.252)
Minimum Wage		-0.179*	0.598	-0.801**
		(0.0859)	(0.872)	(0.369)
R^2		0.543	0.086	0.167
Mean of Dep Variable		5.270	-7.296	0.419
N		1109000	1109000	1109000

Dependent variables are multiplied by 100 (and thus denoted in percentage points). Employment growth measures are defined to equal zero when the firm has no workers of that category. TaxΔ measures the dollar change in the state maximum UI tax from 2009 to the current year. High is an indicator for high-exposure employers. Regressions include firm, NAICS2-year-quarter, and high-year-quarter fixed effects. The necessary DDD interactions are either absorbed by the firm fixed effects or the high-year-quarter fixed effects. Robust standard errors clustered at state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Heterogeneity by Employer Characteristics (2008–2013)

	Establishments			Firm Age			Education		
	Growth	Hires	Sep	Growth	Hires	Sep	Growth	Hires	Sep
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
τ	-0.0599	0.00684	-0.0578	-0.419	-0.174*	-0.115**	1.317***	0.0202	-0.246***
	(0.519)	(0.0966)	(0.0919)	(0.327)	(0.0908)	(0.0483)	(0.418)	(0.0761)	(0.0650)
$\tau \times$ High	-1.307*	-0.0630	0.133	-1.035*	-0.243	0.0846	-2.489***	-0.283*	0.168
	(0.667)	(0.109)	(0.110)	(0.542)	(0.187)	(0.146)	(0.604)	(0.154)	(0.128)
$\tau \times$ Single	1.441***	-0.0166	-0.198***						
	(0.340)	(0.0576)	(0.0553)						
$\tau \times$ High $\times$ Single	-1.303***	-0.330***	0.0310						
	(0.382)	(0.112)	(0.101)						
$\tau \times$ Old				2.315***	0.282**	-0.140**			
				(0.683)	(0.102)	(0.0540)			
$\tau \times$ High $\times$ Old				-1.927***	-0.0218	0.121			
				(0.657)	(0.130)	(0.106)			
$\tau \times$ Low							-0.953*	-0.0675	0.128***
							(0.468)	(0.0801)	(0.0416)
$\tau \times$ High $\times$ Low							0.782	0.0154	-0.0609
							(0.527)	(0.130)	(0.0851)
Minimum Wage	-0.248	-0.123	0.00136	-0.279	-0.126	0.00442	-0.265	-0.122	0.00351
	(0.324)	(0.117)	(0.138)	(0.329)	(0.117)	(0.138)	(0.325)	(0.117)	(0.137)
R^2	0.203	0.475	0.482	0.203	0.475	0.482	0.203	0.475	0.482
Mean of Dep Var	-1.373	10.61	11.28	-1.373	10.61	11.28	-1.373	10.61	11.28
N	1109000	1109000	1109000	1109000	1109000	1109000	1109000	1109000	1109000

Dependent variables are multipled by 100 (and thus denoted in percentage points). τ measures the firm's simulated effective rate increase from 2009 to the current year. High is an indicator for high-exposure employers. Single is an indicator for single-establishment firm (65.8% of sample). Old is an indicator for firm age older than 20 (56.9% of sample). Low is an indicator for more than 40 percent of firm's workforce having no college (40.3% of sample). Regressions include firm, NAICS2-year-quarter, and high-year-quarter fixed effects. Robust standard errors clustered at state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.1 Online Appendix

A.1.1 State UI Tax Increases

This section further describes the quasi-random behavior of states in raising their UI taxes after the Great Recession. State UI trust funds began to run out in 2009, and a total of 34 states resorted to borrowing UI loans from the federal government at some point after the Great Recession.[27] Normally, federal UI loans that are not repaid within a year would begin to accrue interest and subject all employers in the state to a FUTA tax surcharge of $21 per worker at the end of the following year (and multiplies with each additional year of outstanding loans). Thus even without any legislative action by the state, FUTA surcharges ensure that UI loans are eventually repaid.

Due to the severity of the Great Recession, Congress suspended interest payments and FUTA surcharges on federal UI loans until 2011. At the same time, additional legislation – the Unemployment Insurance Solvency Act of 2011 along with Obama's budget proposal – would have potentially raised the minimum required UI tax base for all states from $7,000 to $15,000, and further suspended interest payments until 2013.

While some states proactively chose to replenish their UI trust funds through UI tax increases in 2010 and 2011, others chose to wait it out in hopes of federal UI reform. Indiana's legislators originally passed a UI tax increase that would take effect in January of 2010, but delayed it a year to 2011. As an Indianapolis Business Journal article from the time quotes, *"That's exactly what led Hershman to propose delaying implementation of the bill Indiana passed last year. He said it's impossible to tell right now what will emerge at the national level. On one end, there's the chance Obama could nationalize the system, mutualizing the problem across 50 states [...] On the other end, the unemployment system could be overhauled entirely and made the responsibility of individuals [...] There will come a point in which we'll have to right the system. But there may be significant federal action that gives us a clearer picture on our options for doing so."* (Schnitzler (2009)).

This was also the reason for California's inaction. A report released by California's Legislative Analyst's Office stated: *"It is currently unclear whether any federal reforms will be enacted. This uncertainty complicates the Legislature's decision as to how it should address the insolvency of its UI fund. By acting now, the Legislature could stop the growth of the UI fund deficit and reduce associated state interest costs. On the other hand, such actions have the disadvantage of increasing employment costs and/or decreasing aid to unemployed workers during what is likely to remain a difficult economic time for the state."* (Taylor

[27]The states without federal loans were: AK, DC, IA, LA, ME, MS, MT, ND, NE, NM, OK, OR, TN, UT, WA, WV, WY

(2011))

Finally, some states chose to repay their federal UI loans through alternative financing means. Texas issued debt in the private securities market, and Idaho borrowed from the private bond market (Vroman (2011)). While the federal government creates mandates regarding minimum tax payments, there is currently no solvency requirement for state UI financing. States that borrowed UI loans can slowly pay the loans back as the economy recovers. Thus, many of the states that chose not to increase their UI taxes ended up having outstanding federal UI loans that were not fully repaid until 2012 or 2013 (as was the case in the control states of Delaware, Florida, and Missouri). Most notably, it took California until 2017 to pay back all of its program loans.

Appendix Figure A.7 also shows that there was no correlation between UI tax increases and Democratic vote shares in the 2008 presidential election.

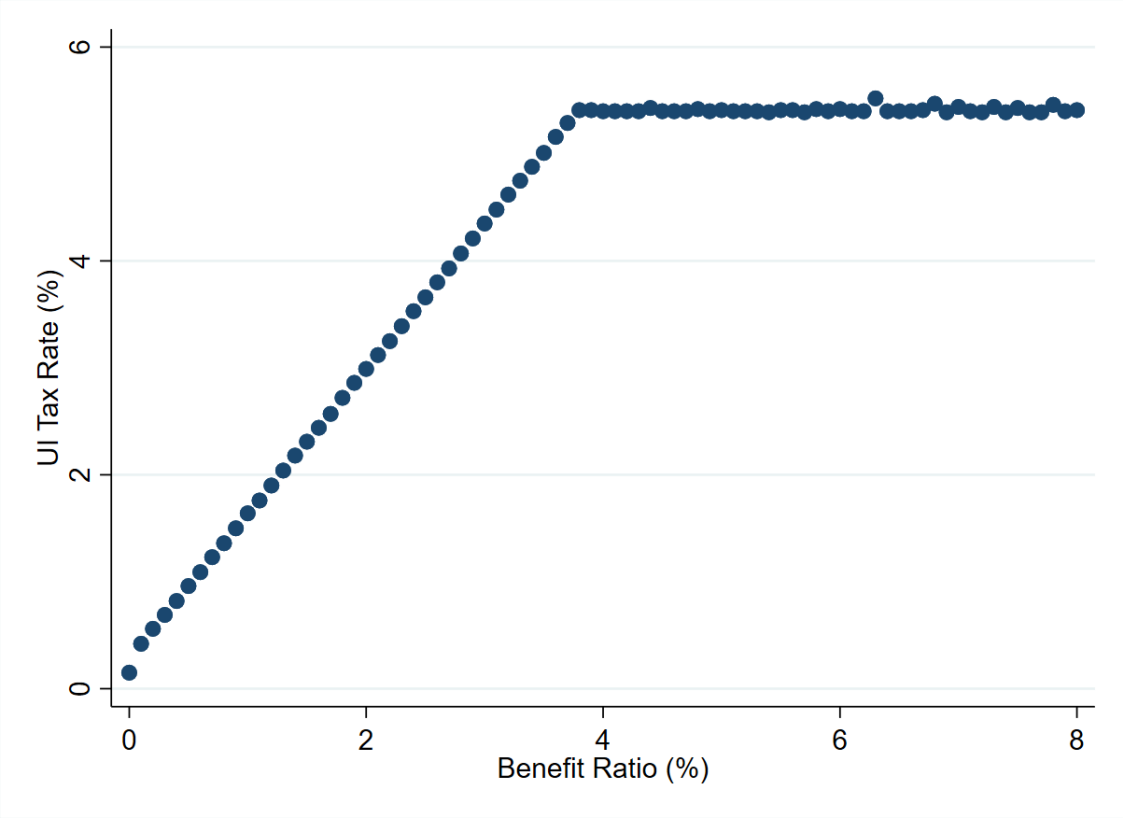
A.1.2 Imputing UI Tax Exposure

It is not layoffs themselves that impact employer-specific tax rates, but rather the dollar amount of UI benefits claimed by laid off workers, relative to total taxable payroll. In order to identify separations that could be eligible for UI benefits, I first limit the scope of consideration to workers with at least 2 consecutive quarters of earnings, and high quarter earnings of at least \$1500 (the minimum requirement to be eligible for UI benefits in most states). I also ignore workers younger than age 18 as those job separations could be driven by schooling, and ignore workers older than age 60 as those separations could be driven by retirement. I then impute a layoff when I observe a job separation in quarter t that results in at least one quarter of zero earnings from quarters $t+1$ to $t+3$. Separations that were the result of moves to a different state will not be captured in the LEHD data, so these could also be miscoded as layoffs.

This method is still an imperfect predictor of eventual experience rating, because of imperfect take-up rates and substantial variation in UI duration. In new work using administrative data from Washington, Lachowska et al. (2023) document significant employer-level heterogeneity in the take-up of UI benefits. I view this exercise as accepting there will be false positives (separations that do not result in UI charges), in favor of minimizing false negatives (missing potential UI charges). Thus the measurement error introduced by inadvertently including some employers with low UI tax exposure in the high-exposure group is likely to attenuate my estimates.

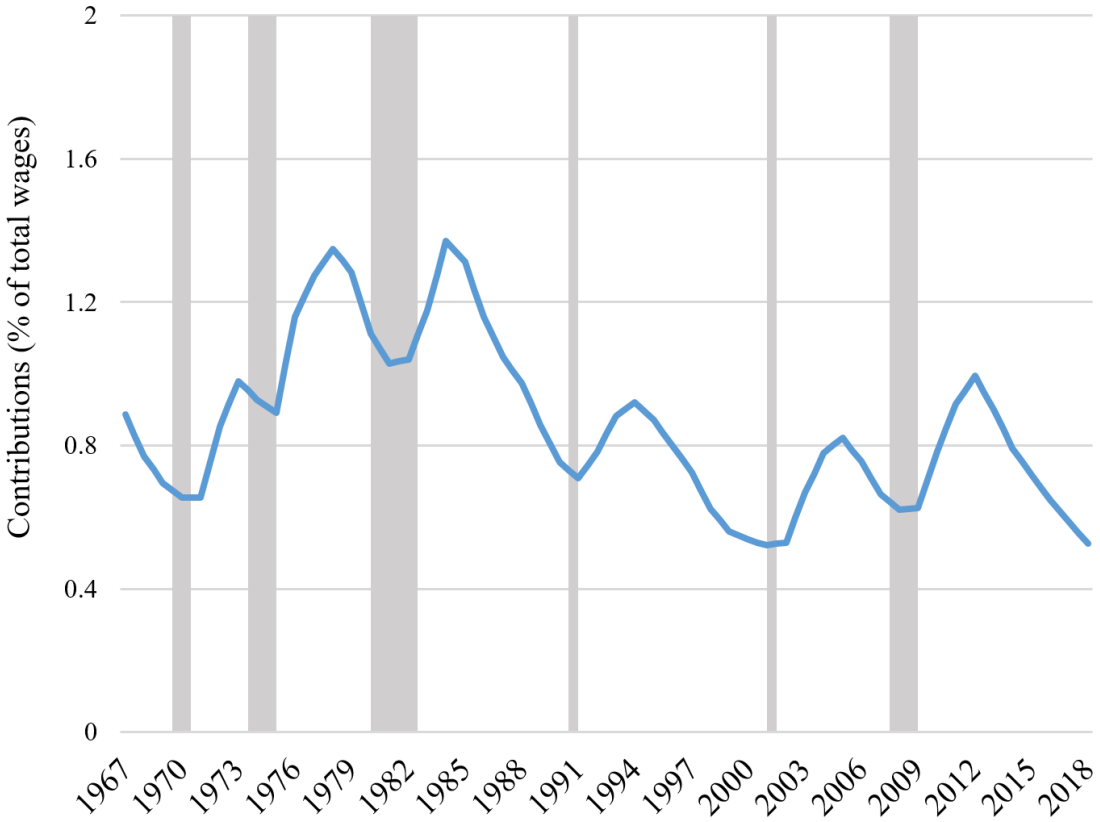
A.2 Appendix Figures and Tables

Figure A.1: Empirical UI Tax Schedule for Florida (2008)



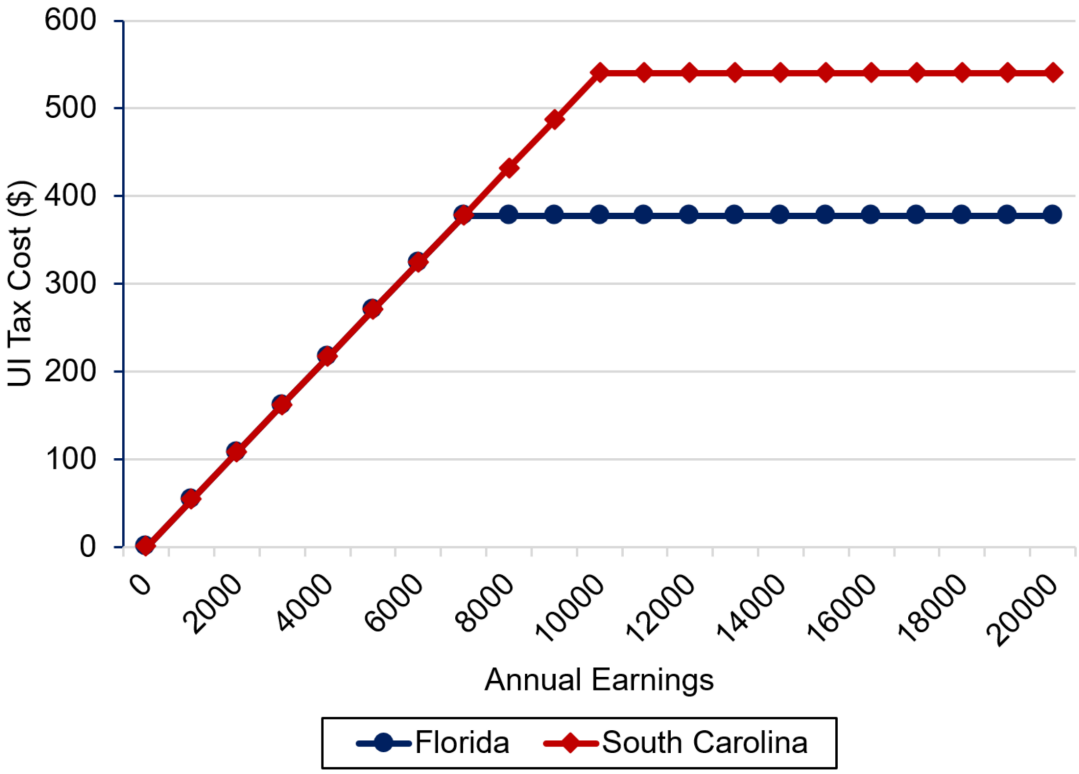
Source: US Dept of Labor ETA 204 Experience Rating Report. The Benefit Ratio is a measure of the employer's layoff experience in the last three years.

Figure A.2: Total UI Contributions, as % of total wages (1967 - 2018)



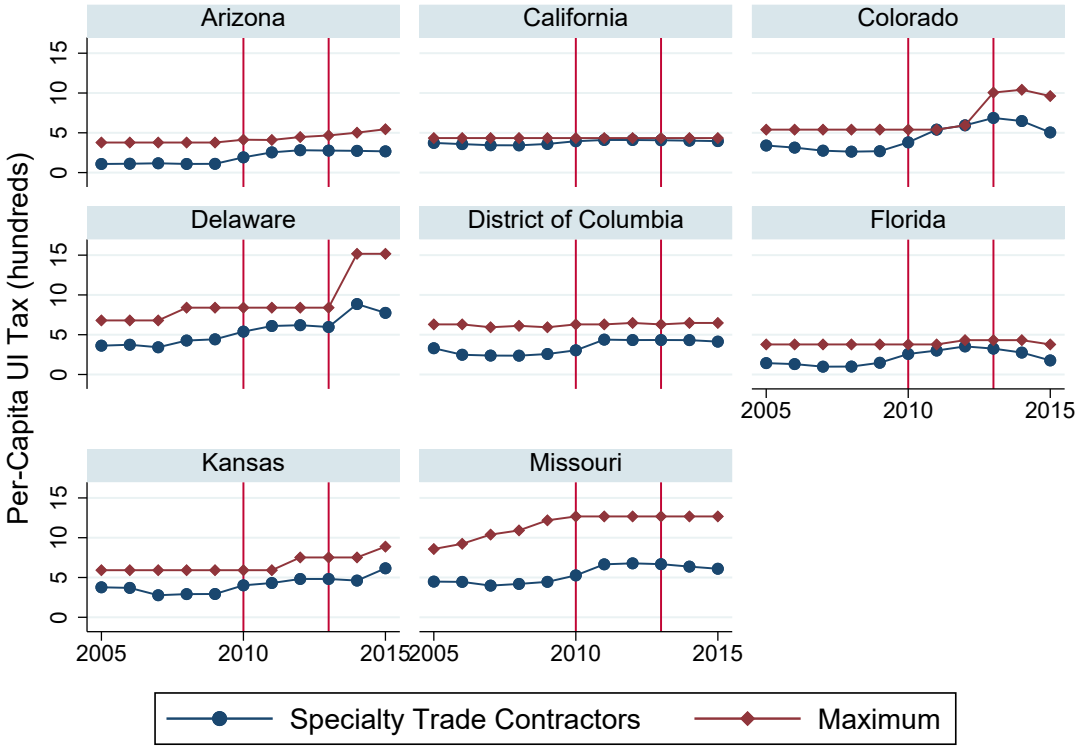
Source: US Dept of Labor Unemployment Insurance Chartbook.
Shaded regions denote US recession years, as defined by NBER.

Figure A.3: UI Tax Costs as a Function of Earnings (2011)



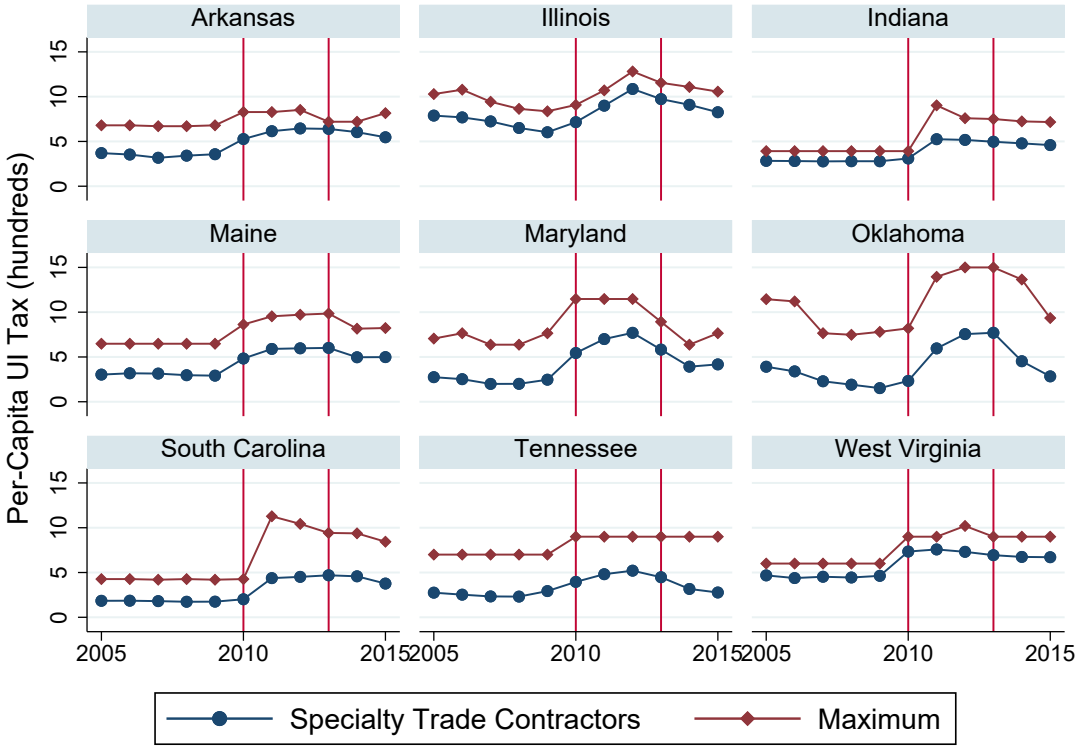
UI tax cost calculated based on a 5.4% tax rate, on a $7,000 tax base in Florida and a $10,000 tax base in South Carolina.

Figure A.4: State UI Tax Schedules - Control



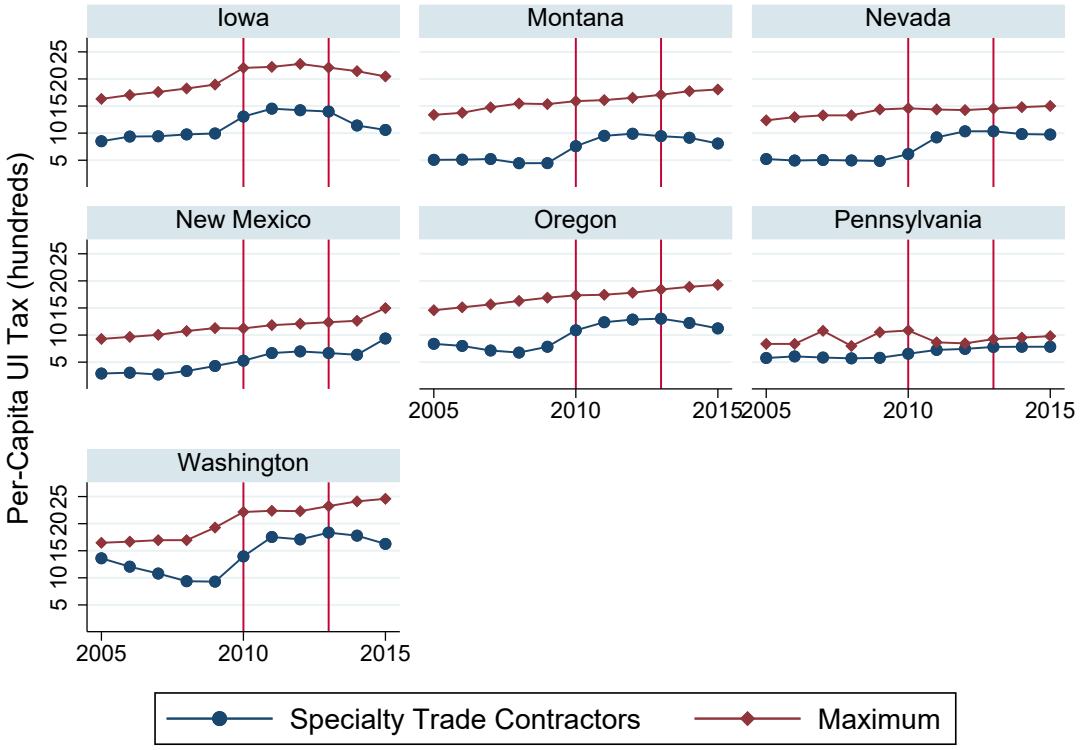
Compares average UI taxes paid and maximum UI taxes. Average taxes for specialty trade contractors obtained from Quarterly Census of Employment and Wages. Specialty trade contractors (NAICS 238) was chosen as an industry whose employers often face rates close to the maximum, to proxy for high exposure firms.

Figure A.5: State UI Tax Schedules - Treatment



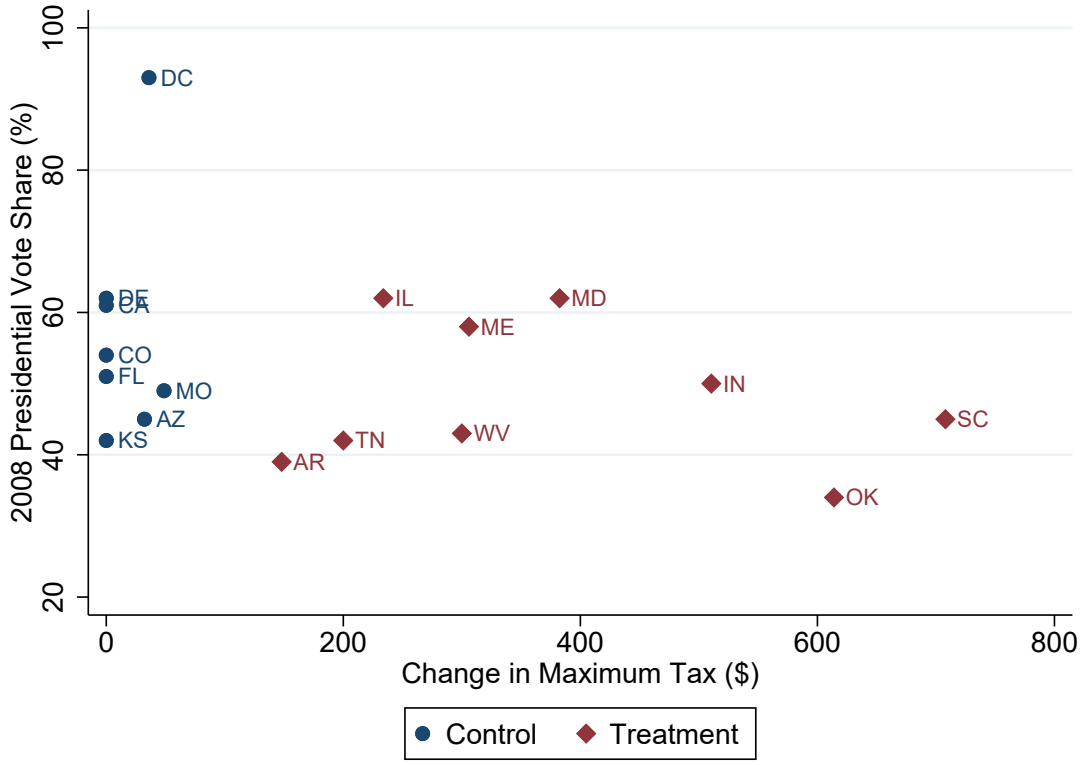
Compares average UI taxes paid and maximum UI taxes. Average taxes for specialty trade contractors obtained from Quarterly Census of Employment and Wages. Specialty trade contractors (NAICS 238) was chosen as an industry whose employers often face rates close to the maximum, to proxy for high exposure firms.

Figure A.6: State UI Tax Schedules - Excluded



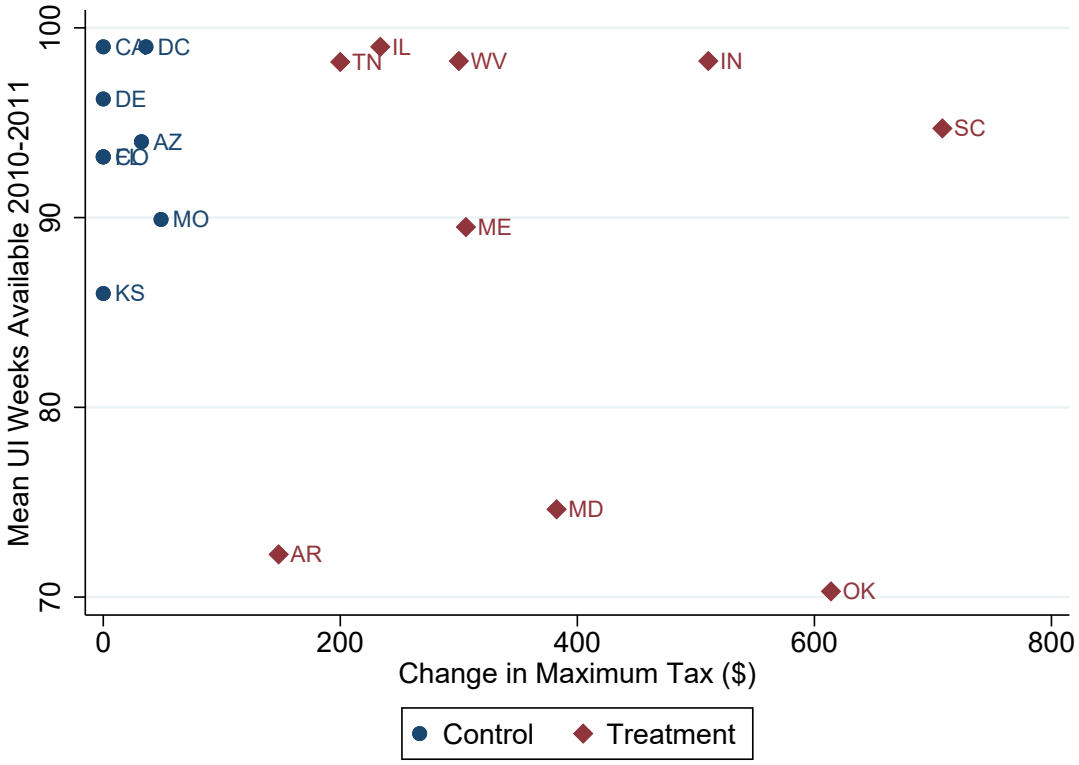
Compares average UI taxes paid and maximum UI taxes. Average taxes for specialty trade contractors obtained from Quarterly Census of Employment and Wages. Specialty trade contractors (NAICS 238) was chosen as an industry whose employers often face rates close to the maximum, to proxy for high exposure firms.

Figure A.7: No Correlation between UI Tax Increases and Presidential Vote Share



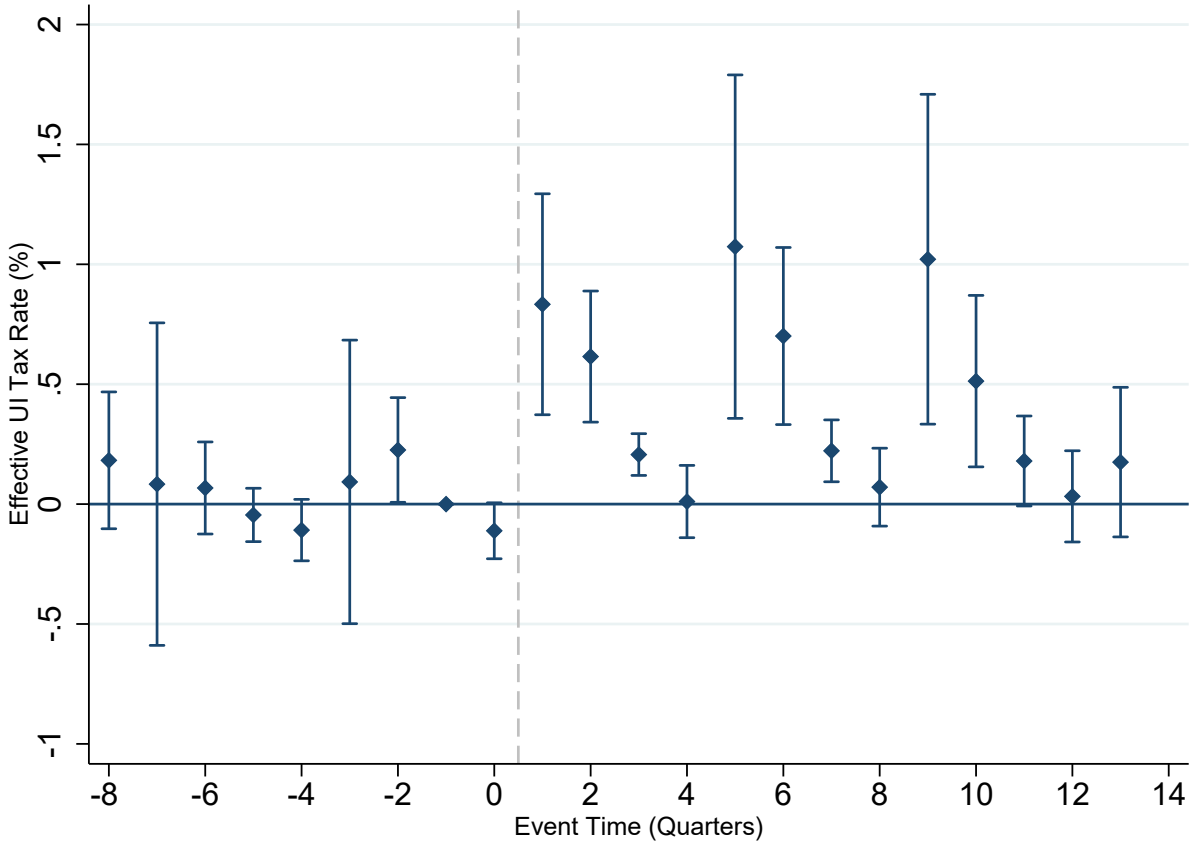
This figure plots the change in the maximum tax from 2009 to 2011 against the state's Democrat vote share in the 2008 presidential election. Vote share data comes from The American Presidency Project (`https://www.presidency.ucsb.edu/statistics/elections/2008`)

Figure A.8: No Correlation between UI Tax Increases and Extended UI Benefits



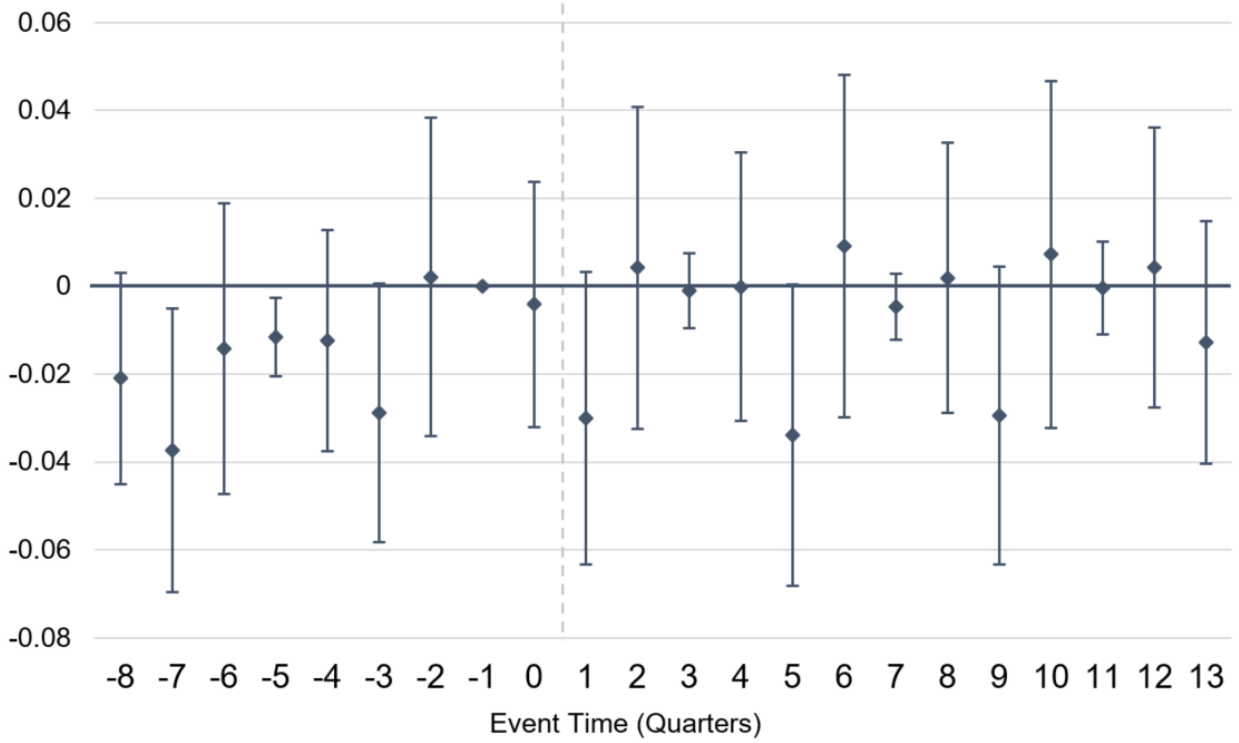
This figure plots the change in the maximum tax from 2009 to 2011 against the state's mean number of benefit weeks available from 2010 to 2011. Extended benefits data comes from Farber et al. (2015).

Figure A.9: Effective Quarterly UI Tax Rates (2008-2013)



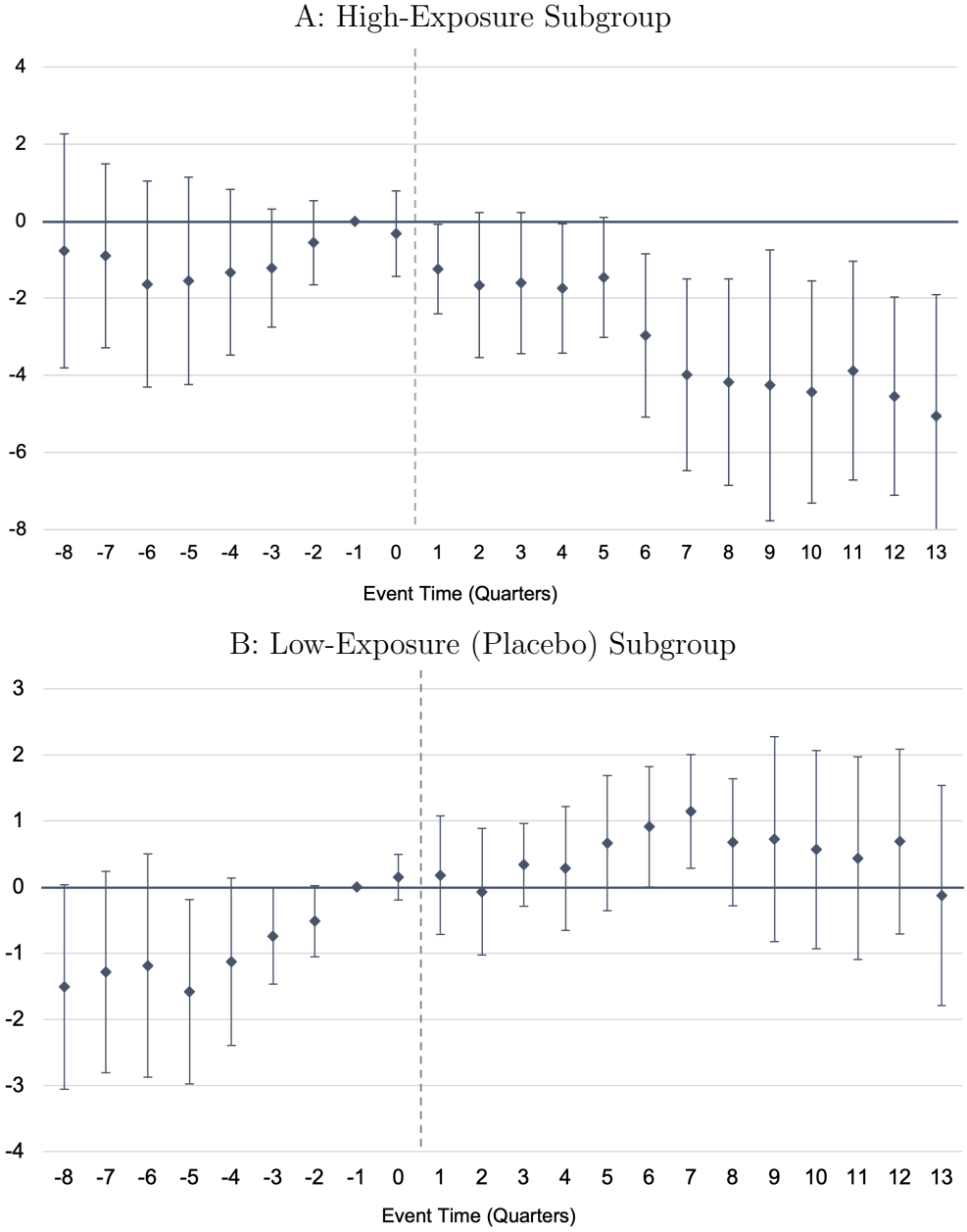
Source: *Quarterly Census of Employment and Wages*. (N = 4,062) Sample limited to ten 4-digit construction industries and years 2008 to 2013 (cells too small to meet disclosure requirements are missing). Effective tax rates calculated by dividing quarterly UI contributions by quarterly payroll. Event study estimated with state-industry and year-quarter fixed effects, and weighted by industry employment. Error bars denote 95% CI for standard errors clustered at state level.

Figure A.10: Log Quarterly Earnings - Triple Difference (2008-2013)



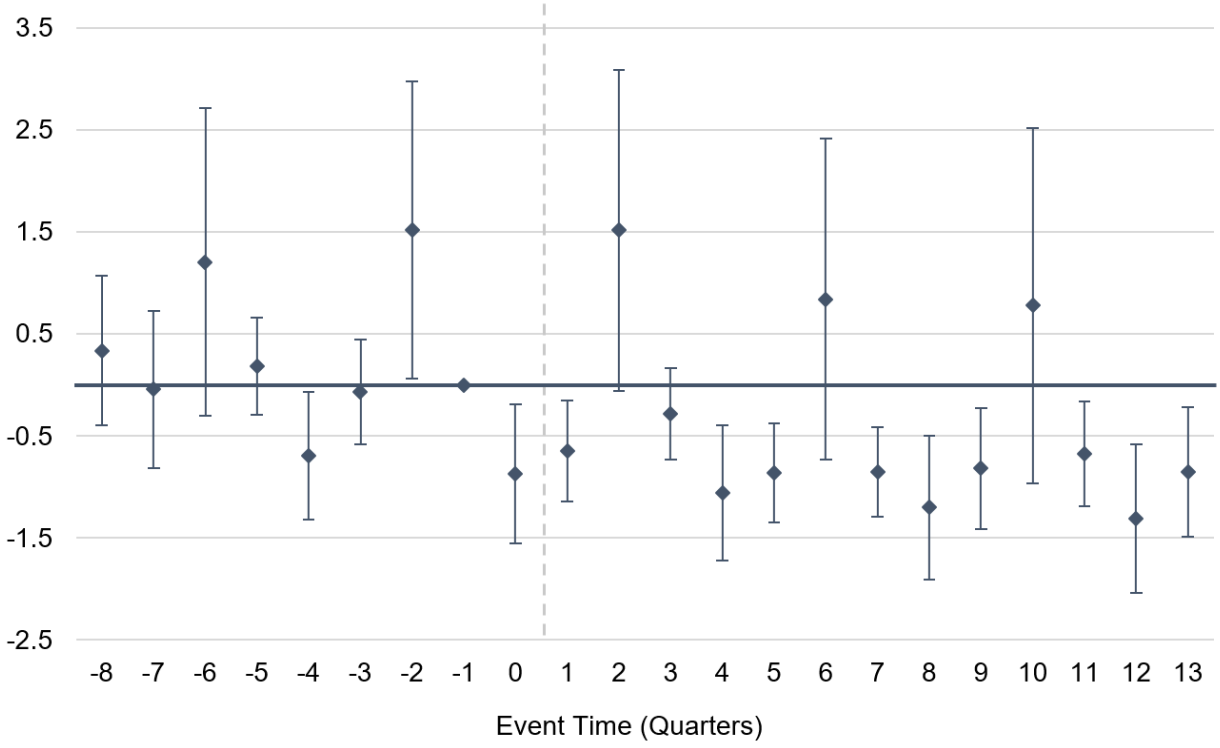
N = 1,109,000. Outcome variable is the mean quarterly earnings of stable workers (employed in both t-1 and t+1). Figure plots the estimates from treatment dummies interacted with event time and high-exposure status. Regression also includes treatment interacted with event time, a control for state minimum wage, and firm and Year-Quarter-NAICS2-High fixed effects. Error bars denote 95% CI for robust standard errors clustered at state level.

Figure A.11: Difference-in-Differences Event Study of Employment Growth (2008-2013)



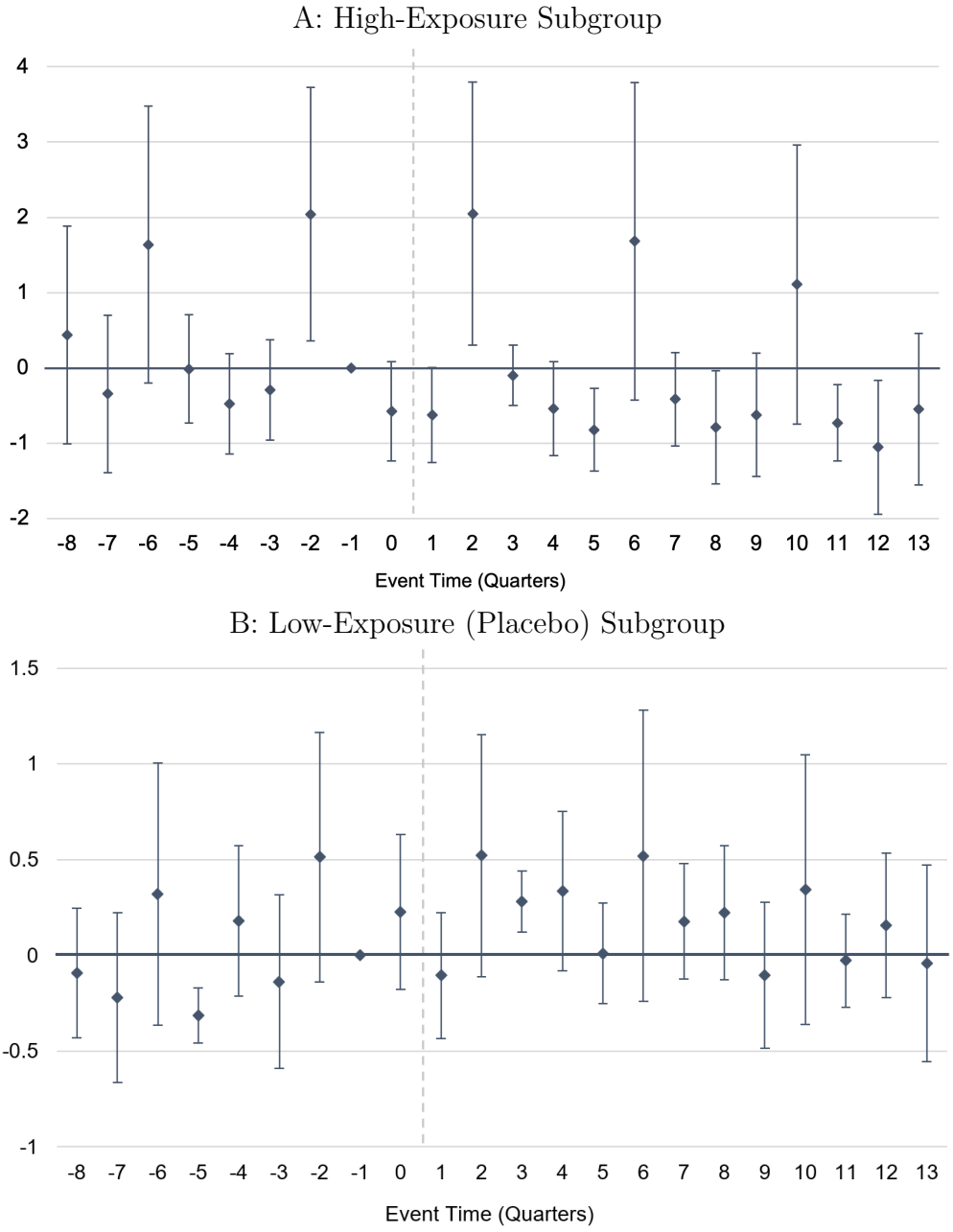
N = 651,000 and 458,000. Outcome variable is the year-over-year employment growth rate. Figure plots the estimates from treatment dummies interacted with event time, for separate subgroups of high-exposure and low-exposure employers. Regression also includes a control for state minimum wage, firm fixed effects, and NAICS2-year-quarter fixed effects. Error bars denote 95% CI for robust standard errors clustered at state level.

Figure A.12: Triple-Difference Event Study Estimates of New Hire Share (2008-2013)



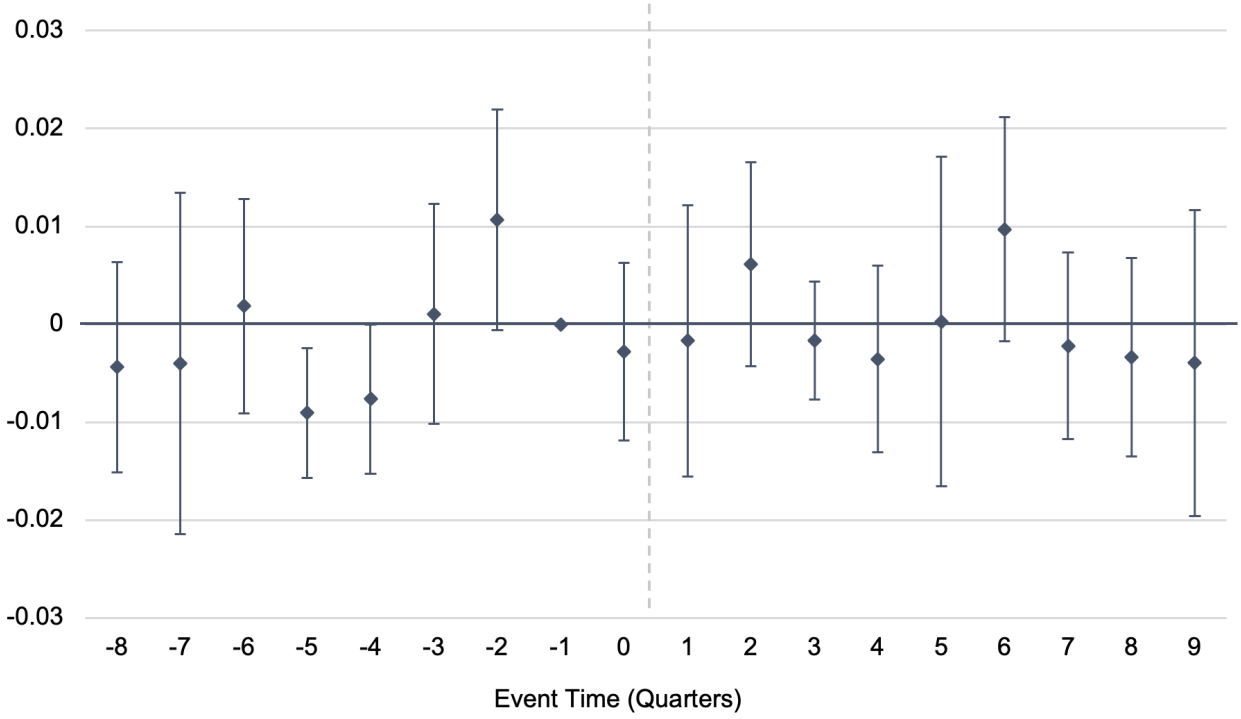
N = 1,109,000. Outcome variable is quarterly new hires divided by employment (expressed in percentage points). Figure plots the estimates from treatment dummies interacted with event time and high-exposure status. Regression also includes treatment dummies interacted with event time, a control for state minimum wage, and fixed effects for firm, NAICS2-year-quarter, and high-year-quarter. Error bars denote 95% CI for robust standard errors clustered at state level.

Figure A.13: Difference-in-Differences Event Study of New Hire Share (2008-2013)



N = 651,000 and 458,000. Outcome variable is quarterly new hires divided by employment (expressed in percentage points). Figure plots the estimates from treatment dummies interacted with event time, for separate subgroups of high-exposure and low-exposure employers. Regression also includes a control for state minimum wage, firm fixed effects, and NAICS2-year-quarter fixed effects. Error bars denote 95% CI for robust standard errors clustered at state level.

Figure A.14: Log Earnings - Weighted Worker Level (2008-2012)



N = 17,190,000. Includes 1,401,000 unique workers across 27,500 unique employers. Figure plots the estimates from ratechange measure interacted with event time. Observations are weighted by 1 divided by baseline (2009:Q3) firm employment. Regression also includes a control for state minimum wage, lagged log earnings, and firm, year-quarter, and state-by-calendar quarter fixed effects. Error bars denote 95% CI for robust standard errors clustered at state level.

Table A.1: Pooled Triple Difference Estimates - Weighted by Employment (2008–2013)

	Employment Growth	New Hire Share	Separations Share	Log Earnings
	(1)	(2)	(3)	(4)
$\tau \times$ Q1	0.759	-0.612***	-0.514***	-0.0101
	(0.629)	(0.141)	(0.172)	(0.00610)
$\tau \times$ Q2	0.794	0.247	-0.0617	0.0187
	(0.578)	(0.202)	(0.308)	(0.0141)
$\tau \times$ Q3	0.669	0.119	-0.263	0.00356
	(0.609)	(0.139)	(0.272)	(0.0189)
$\tau \times$ Q4	0.451	0.0504	-0.0477	-0.00102
	(0.615)	(0.0773)	(0.287)	(0.00641)
$\tau \times$ Q1 $\times$ High	-2.245**	-0.0102	0.364***	0.00619
	(0.776)	(0.169)	(0.112)	(0.00524)
$\tau \times$ Q2 $\times$ High	-2.696***	0.349	-0.0489	-0.00380
	(0.745)	(0.277)	(0.255)	(0.0140)
$\tau \times$ Q3 $\times$ High	-2.860***	-0.489***	0.751***	0.00987
	(0.826)	(0.158)	(0.240)	(0.0196)
$\tau \times$ Q4 $\times$ High	-2.732***	-0.680**	0.343	0.00756
	(0.932)	(0.271)	(0.366)	(0.00484)
Minimum Wage	-0.448	-0.0603	0.0385	0.00568*
	(0.432)	(0.129)	(0.142)	(0.00318)
R^2	0.255	0.579	0.605	0.918
Mean of Dep Variable	2.392	11.47	11.72	9.339
N	1109000	1109000	1109000	1109000

Dependent variables are multipled by 100 (and thus denoted in percentage points), and regressions are weighted by employment. τ measures the firm's simulated effective rate increase from 2009 to the current year. High is an indicator for high-exposure employers. Regressions include firm, NAICS2-year-quarter, and high-year-quarter fixed effects. The necessary DDD interactions are either absorbed by the firm fixed effects or the high-year-quarter fixed effects. Robust standard errors clustered at state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.2: Triple Difference Controlling for State-Time FEs (2008–2013)

	Log Earn	Growth	New Hires	Separations	Gr: Under 25	Gr: No College
	(1)	(2)	(3)	(4)	(5)	(6)
TaxΔ × Q1 × High	-0.00292	-0.528**	-0.154*	-0.0456	-0.665**	-0.551**
	(0.00320)	(0.190)	(0.0739)	(0.0601)	(0.284)	(0.231)
TaxΔ × Q2 × High	0.00184	-0.642***	0.196	-0.0838	-0.717**	-0.640**
	(0.00322)	(0.215)	(0.185)	(0.0528)	(0.277)	(0.275)
TaxΔ × Q3 × High	0.00511**	-0.672***	-0.131**	0.130**	-0.776**	-0.723**
	(0.00239)	(0.218)	(0.0504)	(0.0607)	(0.301)	(0.249)
TaxΔ × Q4 × High	0.00256	-0.737***	0.199*	0.195	-0.939**	-0.754***
	(0.00219)	(0.227)	(0.102)	(0.181)	(0.370)	(0.244)
R^2	0.914	0.205	0.479	0.485	0.087	0.168
Mean of Dep Var	9.096	-1.373	10.61	11.28	-7.296	0.419
N	1109000	1109000	1109000	1109000	1109000	1109000

Dependent variables in columns (2)-(6) denoted in percentage points. Regressions include firm, state-year-quarter, NAICS2-year-quarter, and high-year-quarter fixed effects. The necessary DDD interactions are either absorbed by the firm, state-year-quarter, or high-year-quarter fixed effects. Robust standard errors clustered at state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.3: Difference-in-Difference Estimates (2008–2013)

	Employment Growth		New Hire Share		Separation Share	
	(1)	(2)	(3)	(4)	(5)	(6)
TaxΔ $\times$ Q1 ($100's)	-0.142	-0.469**	-0.170**	-0.163**	-0.0968	-0.0350
	(0.129)	(0.177)	(0.0723)	(0.0672)	(0.0832)	(0.0583)
TaxΔ $\times$ Q2 ($100's)	-0.287**	-0.571**	0.240	0.224	-0.0746	-0.0494
	(0.120)	(0.205)	(0.193)	(0.168)	(0.0897)	(0.0757)
TaxΔ $\times$ Q3 ($100's)	-0.318**	-0.674***	-0.104	-0.0962	0.0514	0.102**
	(0.115)	(0.194)	(0.0752)	(0.0756)	(0.0652)	(0.0410)
TaxΔ $\times$ Q4 ($100's)	-0.431***	-0.759***	-0.170*	-0.178*	0.0587	0.110
	(0.127)	(0.171)	(0.0954)	(0.0879)	(0.169)	(0.147)
Minimum Wage	-0.912	-0.792	-0.330*	-0.196	0.107	0.190
	(0.696)	(0.781)	(0.182)	(0.201)	(0.208)	(0.200)
R^2	0.193	0.254	0.427	0.529	0.405	0.521
Mean of Dep Variable	-4.201	0.349	13.40	14.98	14.74	15.87
Weighting		X		X		X
N	651000	651000	651000	651000	651000	651000

Outcomes denoted in percentage points, and even columns are weighted by employment. Regressions include firm and NAICS2-year-quarter fixed effects. Robust standard errors clustered at state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$